ASSIGNMENTS IN JUNIOR MEASUREMENT

TRIGONOMETRY AND RIGHT-ANGLED TRIANGLES

3

Paul Steenson &
Ian Grobbelaar

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INSTANT LESSONS

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3

Paul Steenson &
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Introduction

This book provides a systematic and comprehensive coverage of selected topics from the junior mathematics curriculum. It is independent of any textbook and is not linked to any syllabus.

Each worksheet follows a format comprising three sections:

Section A consists of a set of introductory questions or review of previous knowledge which can be used as a lesson starter or 'warm up' exercise.

Section B provides skills development or drill with graded questions in the focus area of the worksheet.

Section C provides further practice in the form of different types of questions, a riddle, challenge or extension questions.

Review worksheets are also included in each topic area.

There are minimal examples or explanations as teachers have different methods or working that they would like their students to follow.

Most worksheets could be used by students in a range of years or courses. The teacher may decide to use only one or two sections from a worksheet in a lesson or as a homework assignment. The worksheets are designed to be flexible to meet the needs of the teacher and students.

A time required for students to complete each worksheet is not suggested as this will vary significantly depending on the year level of the students and their level of mathematical skill. The teacher is the best person to make this judgement.

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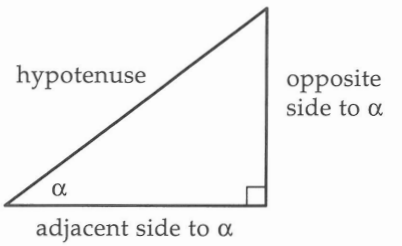
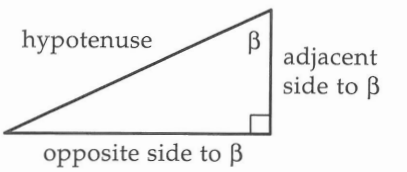
The tangent ratio 1.1

A

Identifying sides

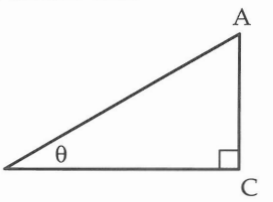
In trigonometry, the sides of a right-angled triangle are given specific names.





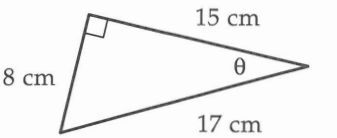
1 Use the diagram to name the:

- **a** hypotenuse
- **b** opposite side
- **c** adjacent side.



2 Write the value of the:

- **a** adjacent side
- **b** opposite side
- **c** hypotenuse.

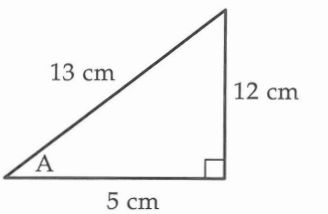


B

The tangent ratio

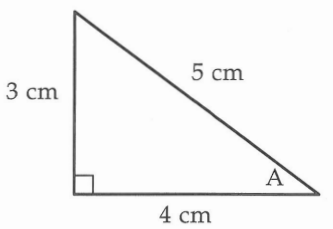
In any right-angled triangle, the tangent ratio is defined:

$$\tan A = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{12}{5}$$



Write the value of tanA for each triangle.

1



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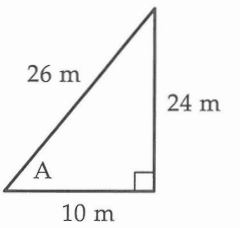

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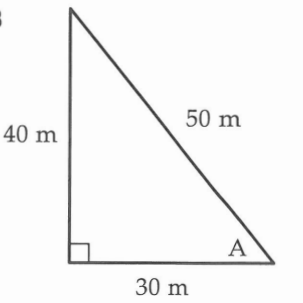

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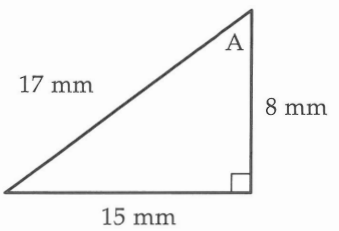

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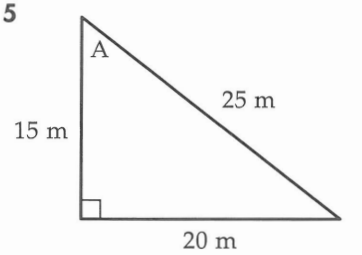

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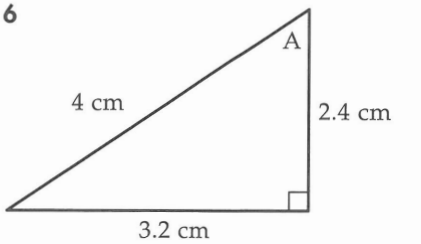

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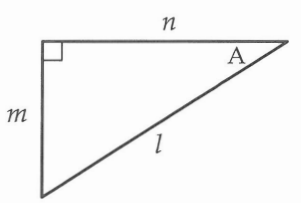

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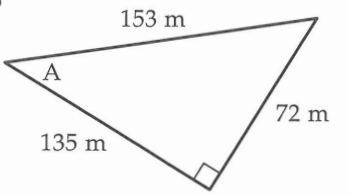

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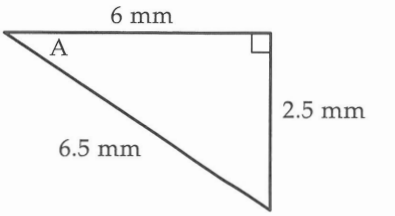

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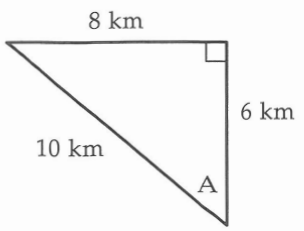
Write the value of tanA for each triangle.

10



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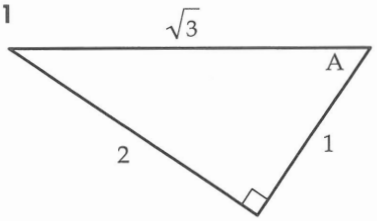

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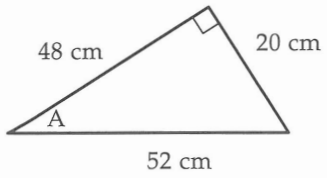

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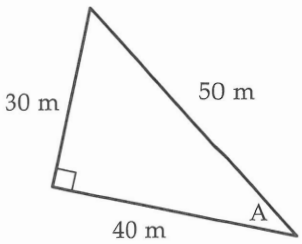

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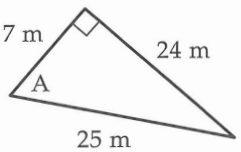

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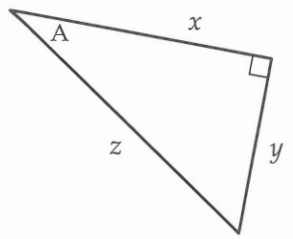
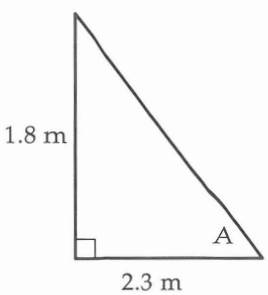


C Use a calculator to find the value of each tan ratio correct to 3 decimal places.

1 tan20°	**2** tan15°	**3** tan55°
4 tan30°	**5** tan43°	**6** tan68°
7 tan75°	**8** tan18°	**9** tan26°
10 tan59°	**11** tan88°	**12** tan33°
13 tan6°	**14** tan17°	**15** tan66°
16 tan51°	**17** tan38°	**18** tan44°
19 tan42°	**20** tan82°	**21** tan21°
22 tan3°	**23** tan19°	**24** tan37°
25 tan9°	**26** tan45°	**27** tan60°
28 tan11°	**29** tan24°	**30** tan10°

Find the value of tanA as a decimal correct to 3 places.

31



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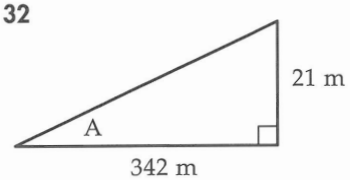

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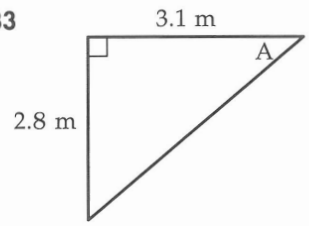

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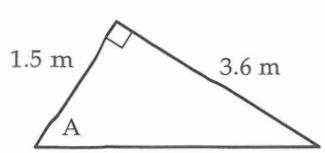

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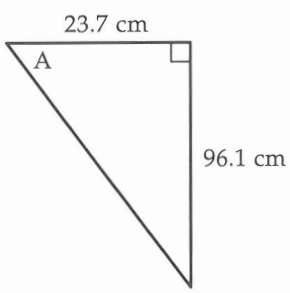

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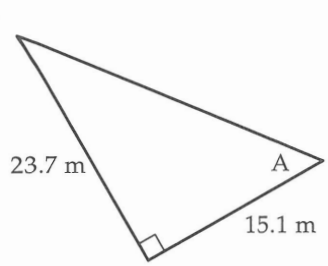

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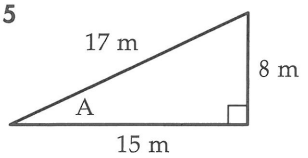
Finding the length of a side *tangent ratio* 2.1

A Find the value of each tan ratio correct to 3 decimal places.

1 tan31° **2** tan15° **3** tan89° **4** tan7°

Find the value of tanA correct to 3 decimal places.

5



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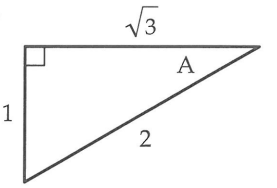

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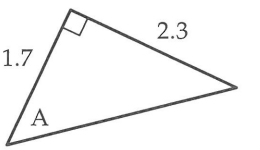

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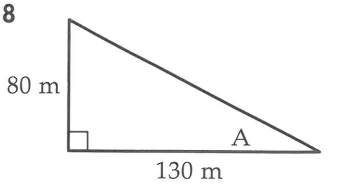
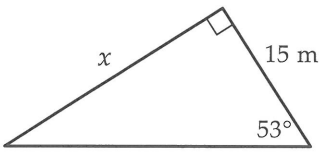


B

The tangent ratio

$$\tan A = \frac{\text{opposite side}}{\text{adjacent side}}$$

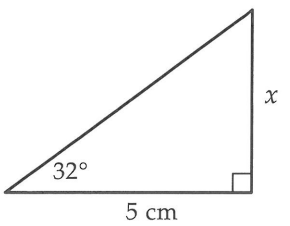
EXAMPLE: Find the length of side x correct to 1 decimal place.



$$\frac{x}{15} = \tan 53°$$
$$x = 15\tan 53°$$
$$= 19.90567232\ldots$$
$$= 19.9 \text{ m}$$

Calculate the length of side x correct to 1 decimal place.

1



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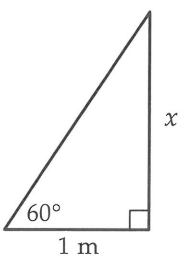

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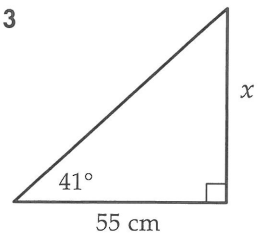

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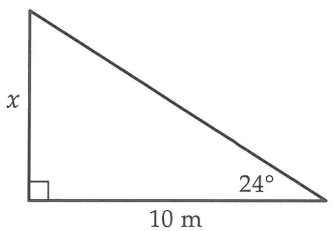

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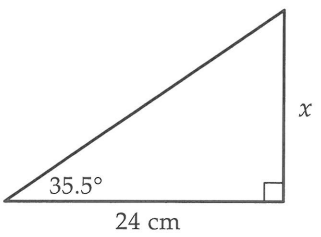

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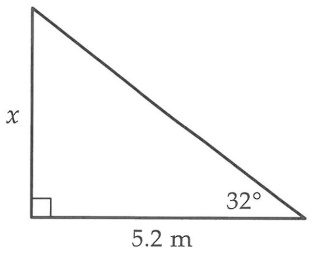

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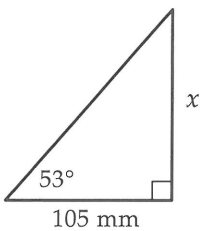

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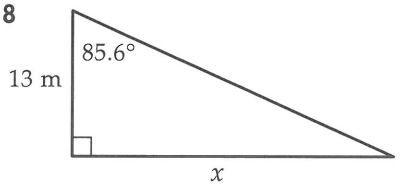

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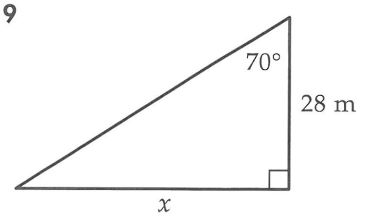

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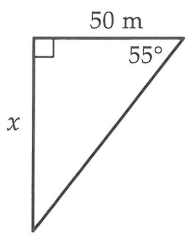

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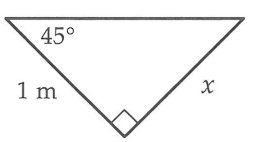

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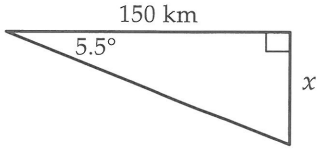

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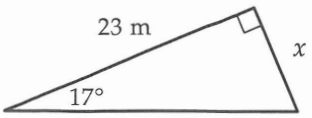
Calculate the length of side x correct to 1 decimal place.

13



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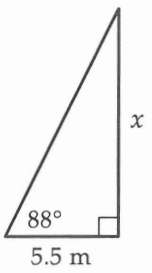

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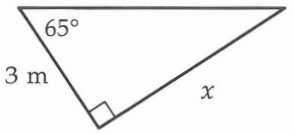

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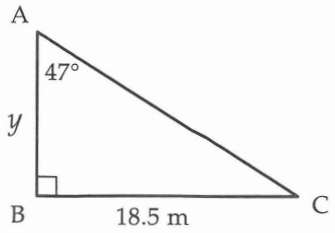


EXAMPLE:

Find the length of side y correct to 1 decimal place.

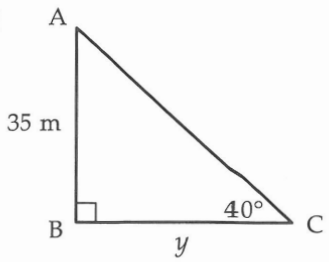
$$\tan 47° = \frac{18.5}{y}$$

$$y = \frac{18.5}{\tan 47°}$$

$$= 17.3 \text{ m}$$

Calculate the length of side y correct to 1 decimal place.

1



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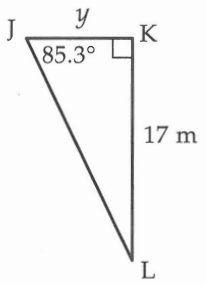

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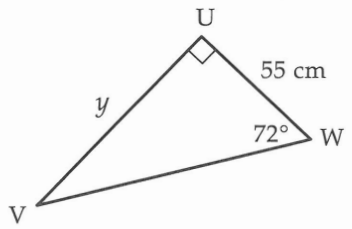

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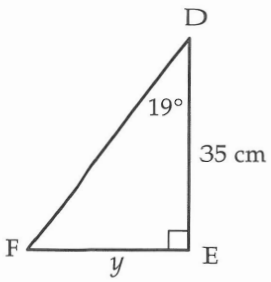

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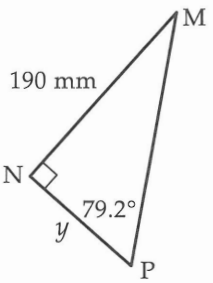

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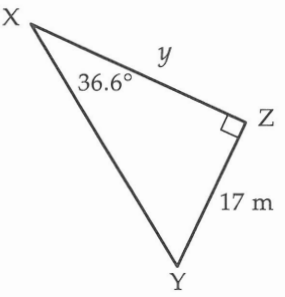

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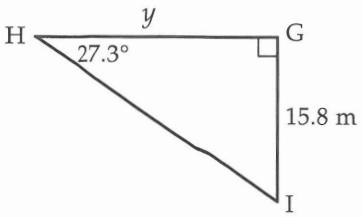

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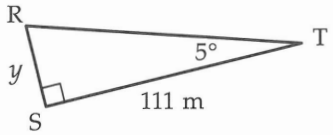

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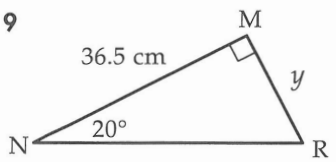

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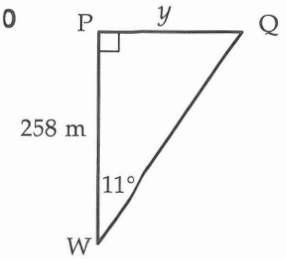

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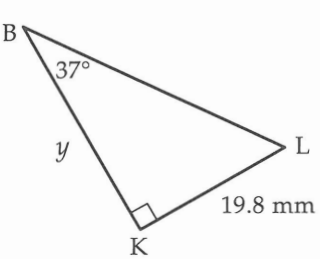

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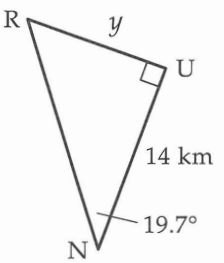

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Note: diagrams not to scale

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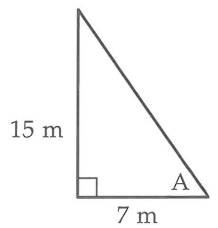
Finding the size of an angle *tangent ratio* 3.1

A Use a calculator to find the size of ∠A to the nearest degree.

1 $\tan A = \frac{4}{5}$
2 $\tan A = 0.65$
3 $\tan A = 1$
4 $\tan A = \frac{3}{4}$
5 $\tan A = 0.02$
6 $\tan A = 27.35$
7 $\tan A = \frac{12}{5}$
8 $\tan A = 1.428$
9 $\tan A = 0.7735$
10 $\tan A = \frac{25}{60}$
11 $\tan A = 57.29$
12 $\tan A = 0.5$

B

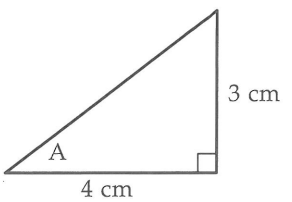


EXAMPLE: Find ∠A to the nearest degree.
The given sides are the opposite and adjacent sides.
Hence: $\tan A = \frac{15}{7}$

$\angle A = 65°$ (to the nearest degree)

Find the size of ∠A correct to 1 decimal place.

1



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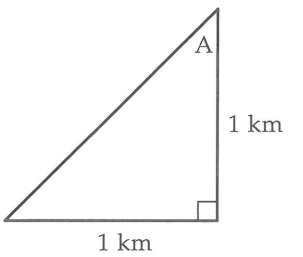

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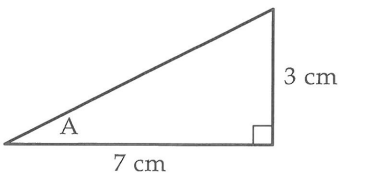

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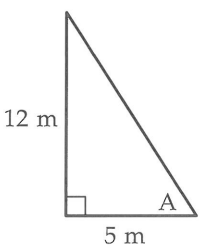

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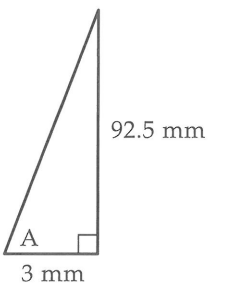

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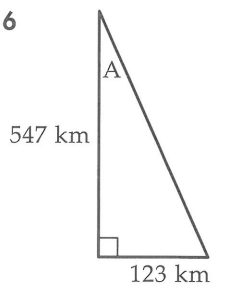

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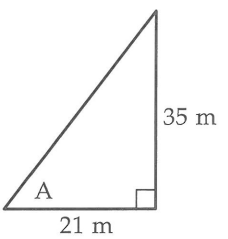

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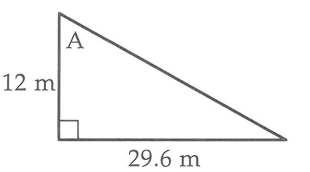

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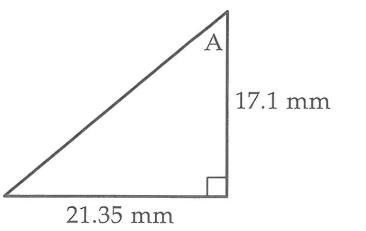

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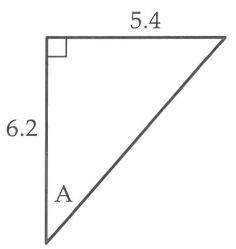

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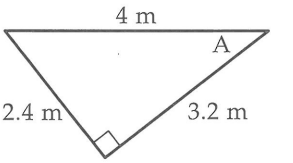

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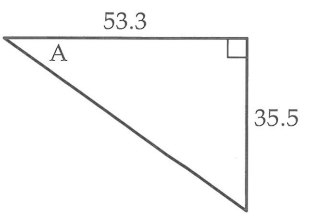

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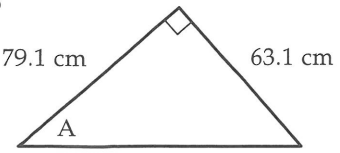
Find the size of ∠A correct to 1 decimal place.

13



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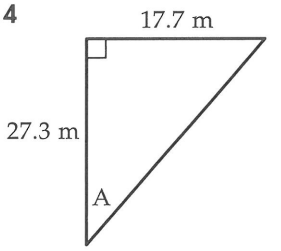

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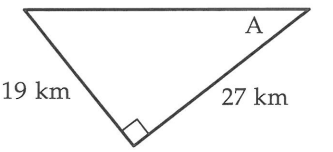
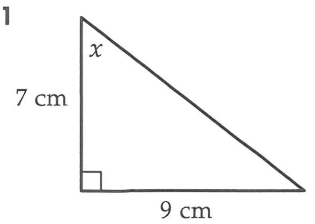


C Find the size of angle x correct to 1 decimal place.

1



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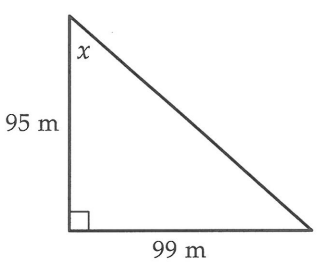

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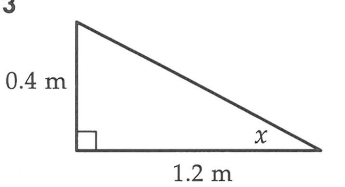

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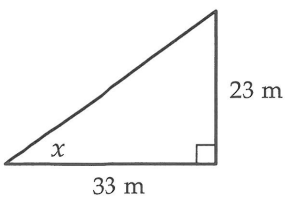

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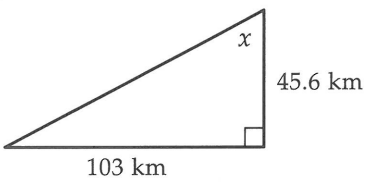

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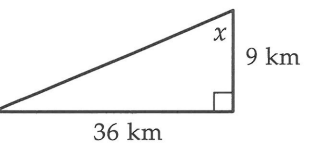

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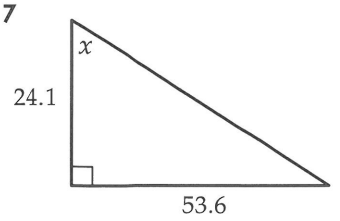

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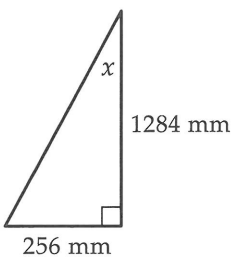

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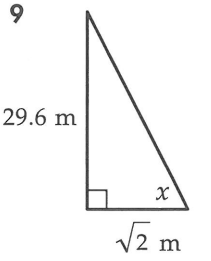

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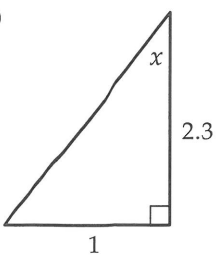

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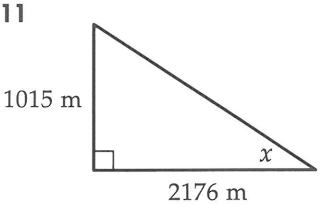

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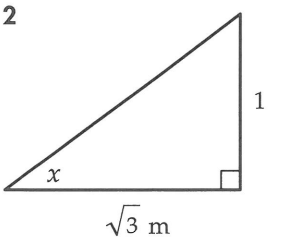

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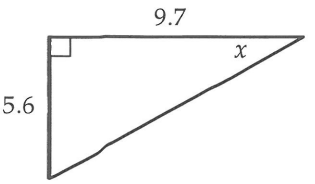

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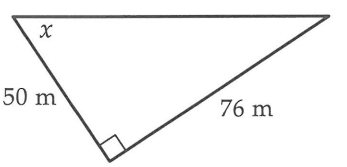

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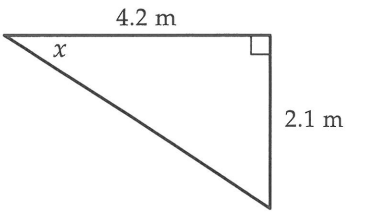

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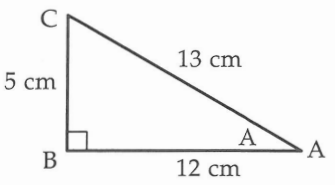
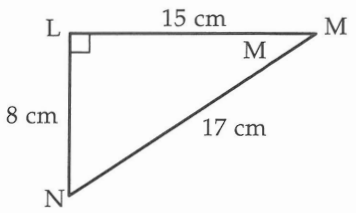
A



For the triangle ABC, give:

1. the length of the hypotenuse.
2. the length of the side opposite ∠A.
3. the length of the side adjacent to ∠A.
4. the value of tan A.



For the triangle LMN, give:

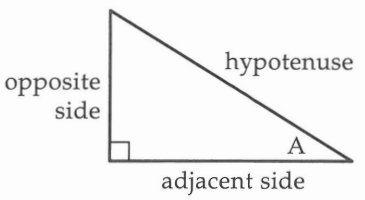
5. the length of the hypotenuse.
6. the length of the side opposite ∠M.
7. the length of the side adjacent to ∠M.
8. the value of tan M.

B

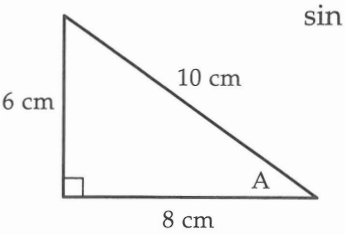
The sine ratio

In any right-angled triangle, the sine ratio is defined:

$$\sin A = \frac{\text{opposite side}}{\text{hypotenuse}}$$



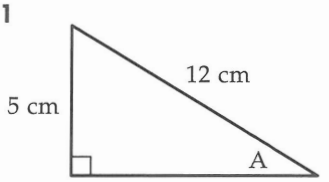
EXAMPLE: Find the value of sin A.



$$\sin A = \frac{6}{10}$$
$$= \frac{3}{5}$$

Calculate the value of sin A for each triangle.

1



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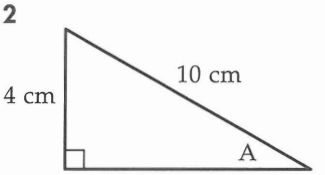

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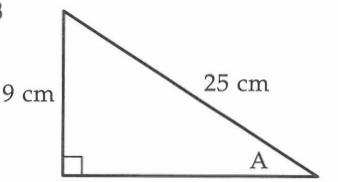

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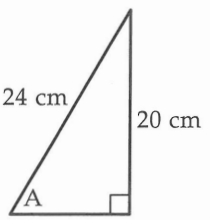

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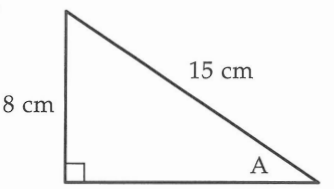

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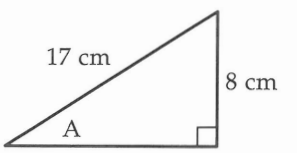

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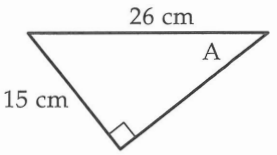

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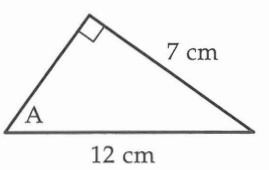

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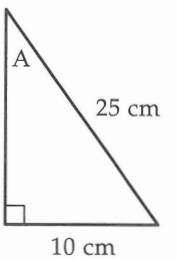

Note: diagrams not to scale

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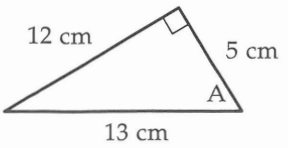
Calculate the value of sin A for each triangle.

10



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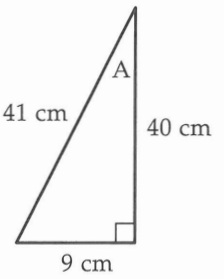

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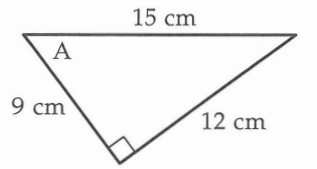

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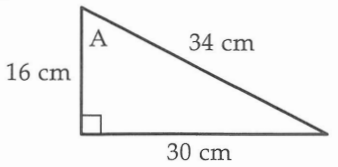

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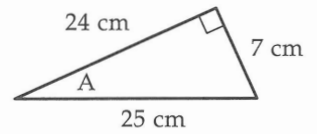

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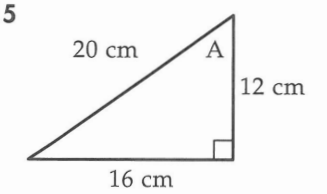


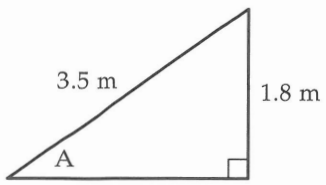
C

Use your calculator to evaluate each ratio correct to 3 decimal places.

1 sin40°	**2** sin25°	**3** sin55°	**4** sin30°
5 sin60°	**6** sin47°	**7** sin68°	**8** sin75°
9 sin18°	**10** sin62°	**11** sin84°	**12** sin36°
13 sin21°	**14** sin73°	**15** sin15°	**16** sin59°
17 sin82°	**18** sin9°	**19** sin39°	**20** sin28°

Find the value of sin A correct to 3 decimal places.

21



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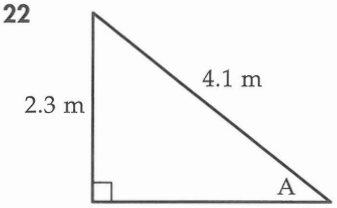

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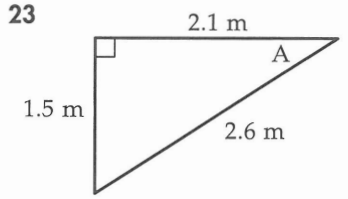

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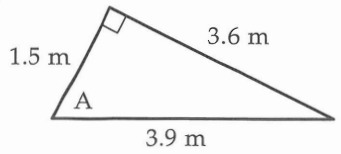

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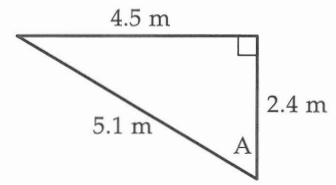

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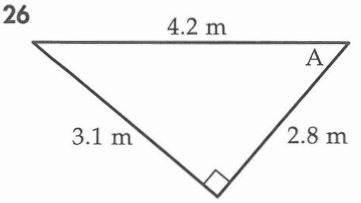

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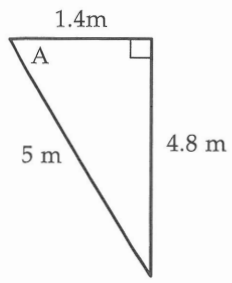

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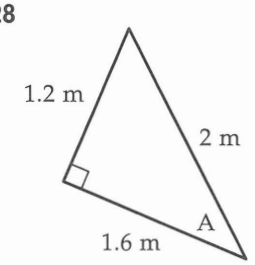

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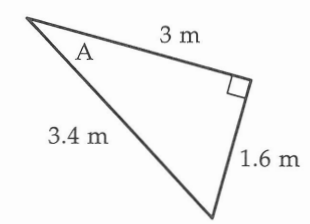

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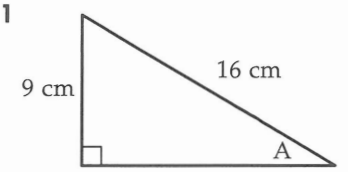
Finding the length of a side *sine ratio* 5.1

A Write down the value of sin A for each triangle.

1



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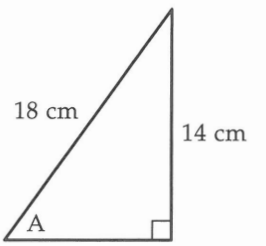

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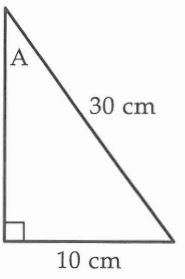

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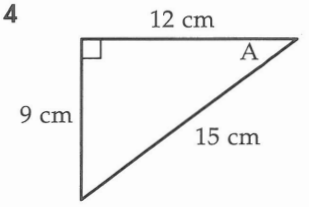

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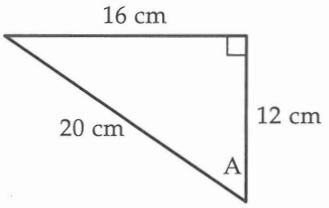

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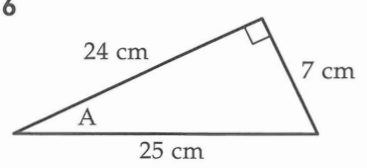


Use your calculator to find the value of each ratio correct to 3 decimal places.

7 sin 42° **8** sin 16° **9** sin 28° **10** sin 65°

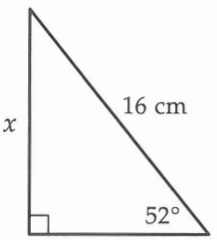
B

The sine ratio

$$\sin A = \frac{\text{opposite side}}{\text{hypotenuse}}$$

EXAMPLE:

Find the value of x correct to 1 decimal place.



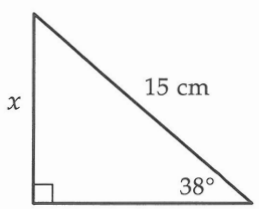
$$\frac{x}{16} = \sin 52°$$

$$x = 16 \sin 52°$$

$$= 12.6 \text{ cm}$$

Use the sine ratio to find the value of x correct to 1 decimal place.

1



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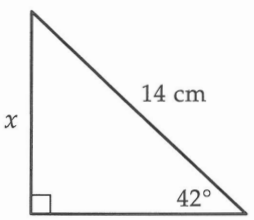

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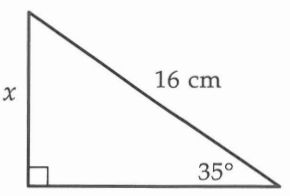

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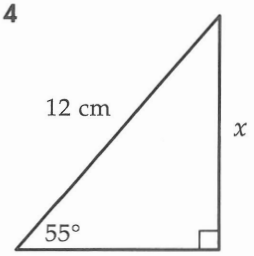

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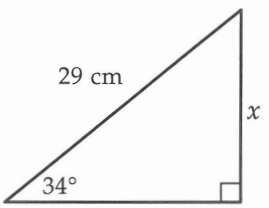

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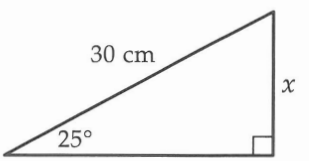

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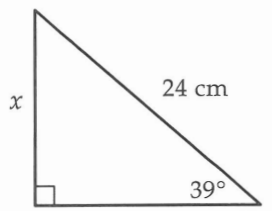
For each diagram, use the sine ratio to find the value of x correct to 1 decimal place.

7



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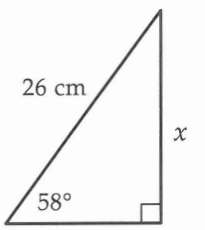

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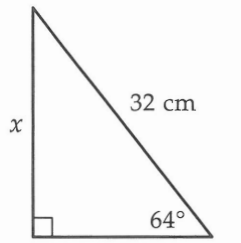

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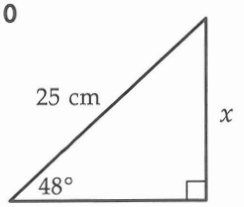

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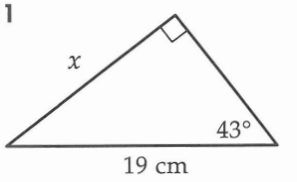

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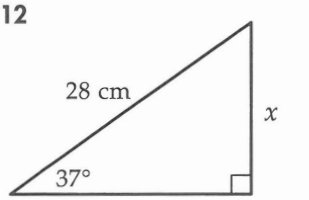

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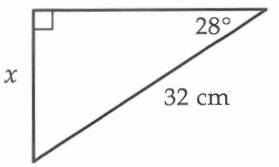

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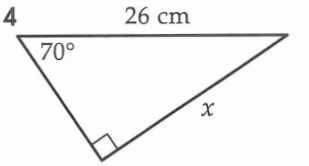

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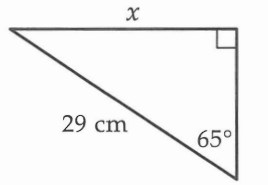

C

For each diagram, calculate the length of side x correct to 1 decimal place.

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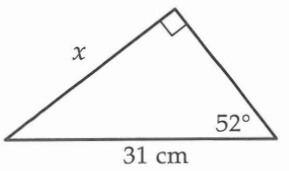

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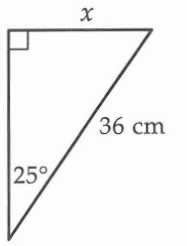

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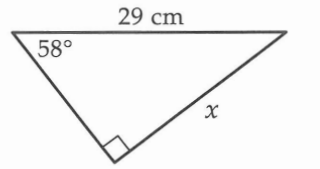

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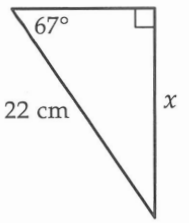

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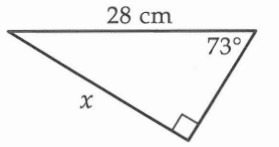

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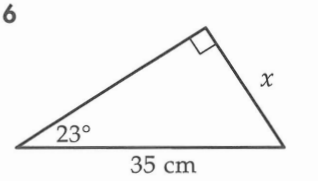

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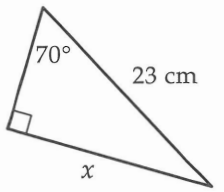

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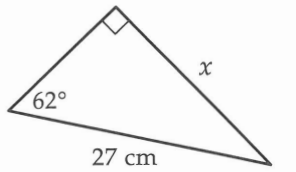

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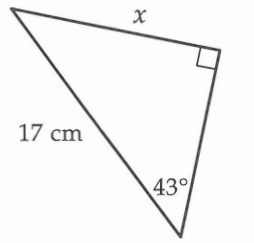

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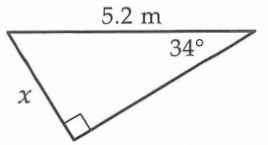

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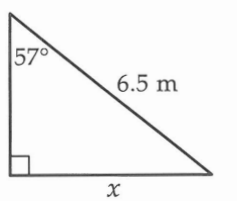

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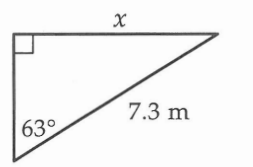

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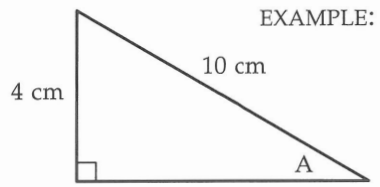
Finding the size of an angle *sine ratio* 6.1

A Use a calculator to find the size of ∠A to the nearest degree.

1 $\sin A = 0.6$
2 $\sin A = 0.8$
3 $\sin A = 0.5$
4 $\sin A = 0.35$
5 $\sin A = 0.92$
6 $\sin A = 0.59$
7 $\sin A = 0.259$
8 $\sin A = 0.861$
9 $\sin A = 0.433$
10 $\sin A = \frac{2}{5}$
11 $\sin A = \frac{3}{4}$
12 $\sin A = \frac{5}{6}$

B

EXAMPLE: Find ∠A to the nearest degree.



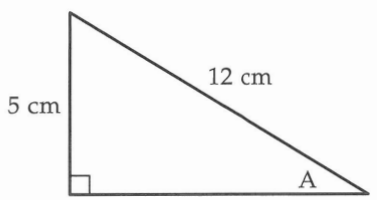
$$\sin A = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$= \frac{4}{10}$$

$$A = 24°$$

Find the size of ∠A to the nearest degree.

1



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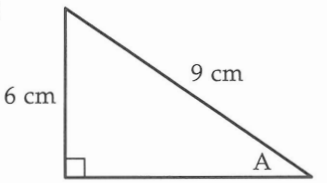

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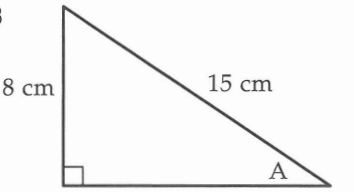

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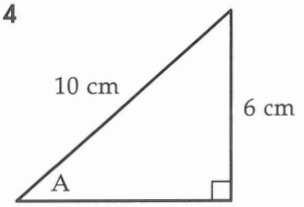

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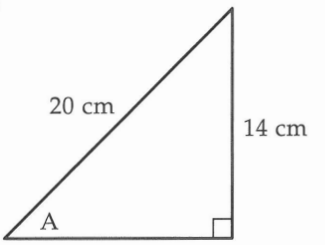

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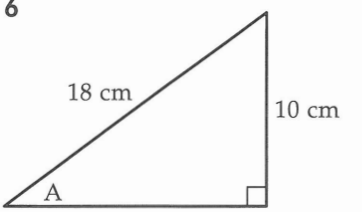

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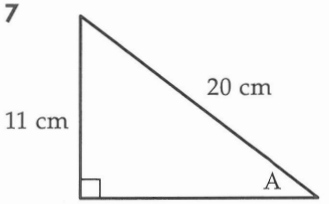

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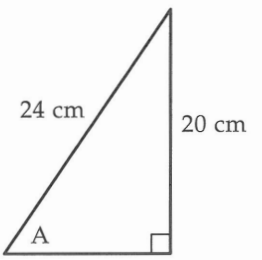

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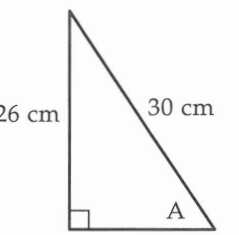

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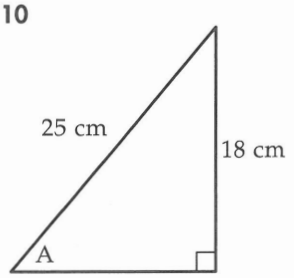

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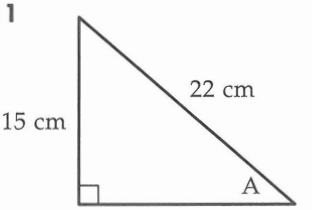

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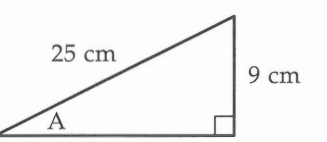

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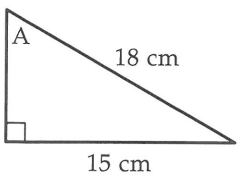
C Find the size of angle A to the nearest degree.

1



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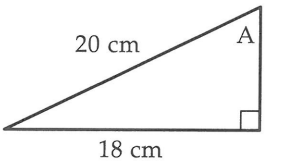

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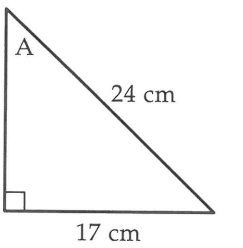

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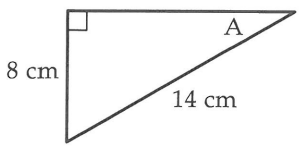

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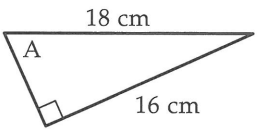

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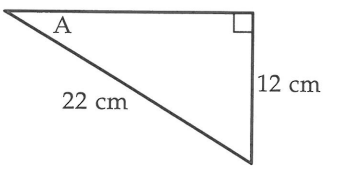

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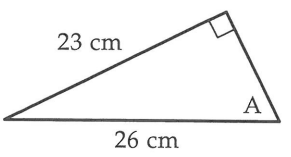

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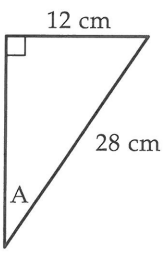

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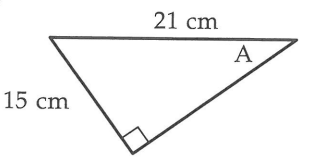

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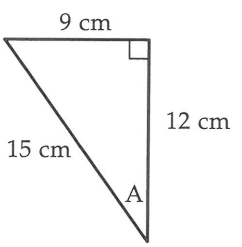

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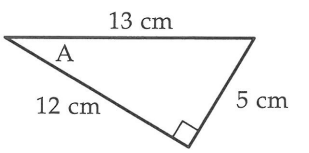

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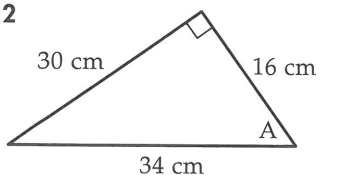

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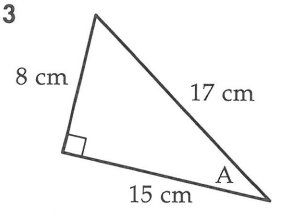

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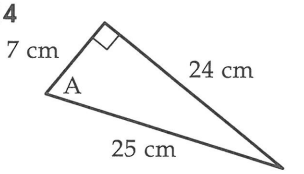

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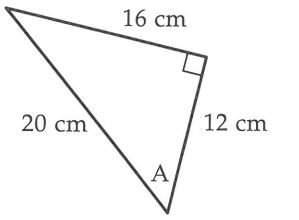

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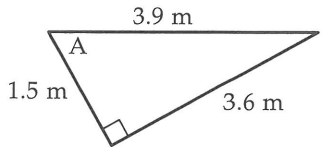

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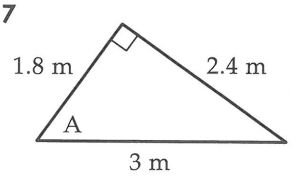

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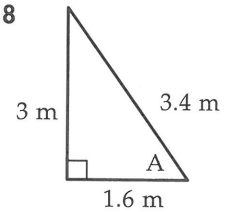

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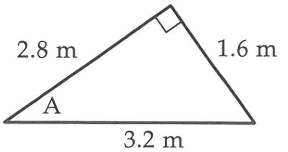

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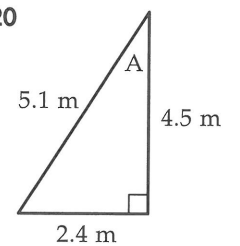

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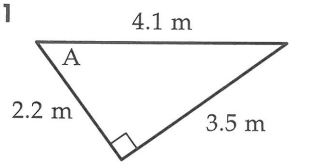

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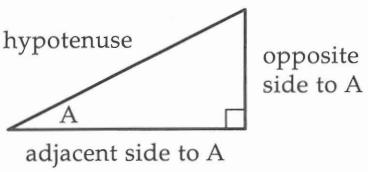
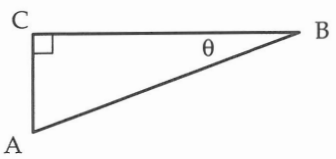
The cosine ratio

A

Identifying sides

Remember, in trigonometry the sides of a right-angled triangle are given specific names.





For the triangle ABC, name:

1. the hypotenuse.
2. the side opposite $\angle\theta$.
3. the side adjacent to $\angle\theta$.

B

The cosine ratio

In any right-angled triangle, the cosine ratio is defined:

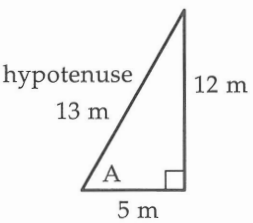
$$\cos A = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

EXAMPLE:

Write the value of $\cos A$ as a simplified fraction.

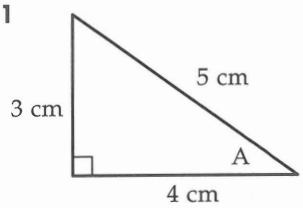
$$\cos A = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$= \frac{5}{13}$$



Find the value of $\cos A$ for each triangle, leaving your answers as simplified fractions.

1



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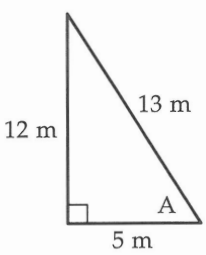

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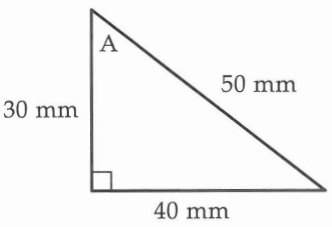

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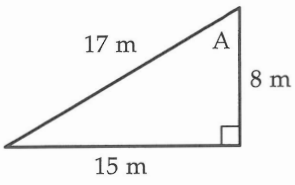

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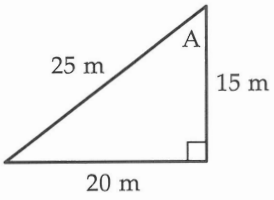

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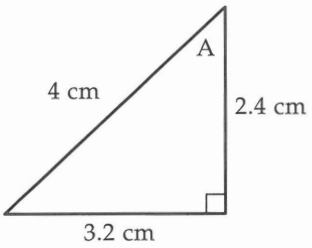

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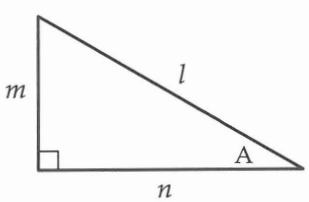

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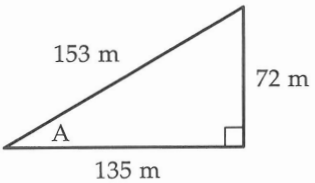

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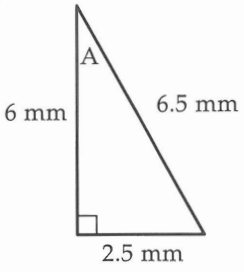

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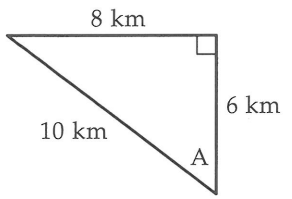
Give the value of cos A for each triangle, leaving your answers as simplified fractions.

10



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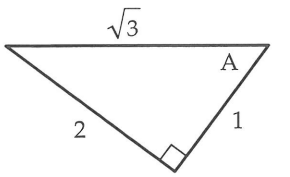

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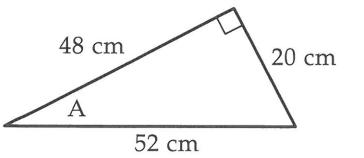

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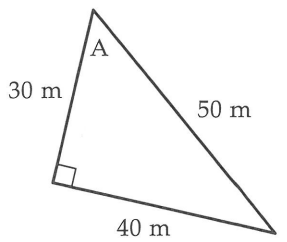

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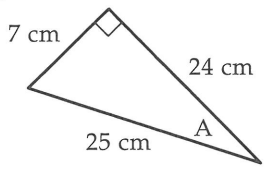

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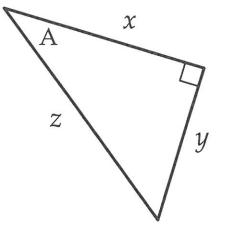

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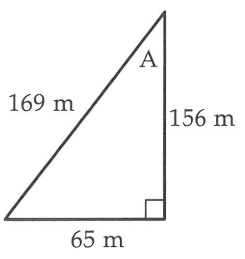

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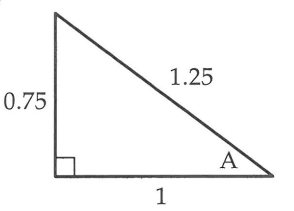

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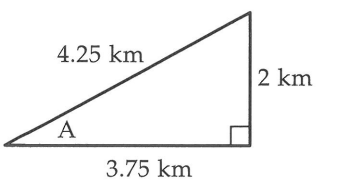
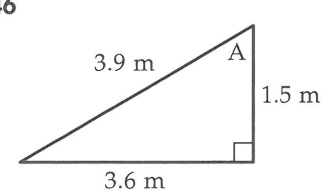


C Use a calculator to evaluate each ratio correct to 3 decimal places.

1 cos 20°	**2** cos 87°	**3** cos 2°
4 cos 81°	**5** cos 44°	**6** cos 12°
7 cos 75°	**8** cos 40°	**9** cos 21°
10 cos 47°	**11** cos 36°	**12** cos 15°
13 cos 55°	**14** cos 45°	**15** cos 66°
16 cos 30°	**17** cos 3°	**18** cos 11°
19 cos 77°	**20** cos 39°	**21** cos 23°
22 cos 19°	**23** cos 60°	**24** cos 82°
25 cos 62°	**26** cos 28°	**27** cos 59°
28 cos 68°	**29** cos 19°	**30** cos 8°
31 cos 4°	**32** cos 76°	**33** cos 53°
34 cos 68°	**35** cos 79°	**36** cos 89°
37 cos 37°	**38** cos 18°	**39** cos 25°
40 cos 84°	**41** cos 73°	**42** cos 9°
43 cos 84°	**44** cos 13°	**45** cos 1°

For each triangle, calculate the value of cos A as a decimal correct to 3 places.

46



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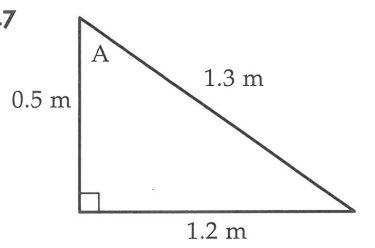

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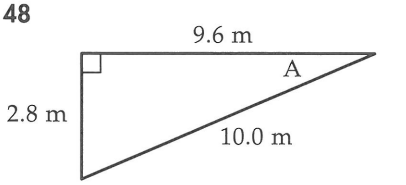

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Note: diagrams not to scale

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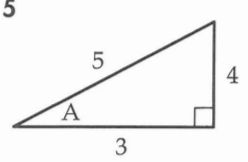
Finding the length of a side *cosine ratio* 8.1

A Find the value of each cos ratio correct to 3 decimal places.

1 $\cos 30°$ **2** $\cos 60°$ **3** $\cos 90°$ **4** $\cos 45°$

Find the value of $\cos A$ correct to 3 decimal places.

5



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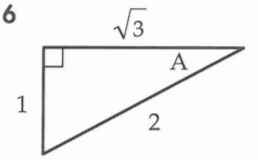

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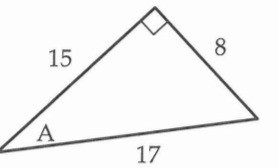

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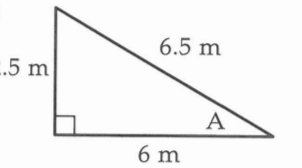


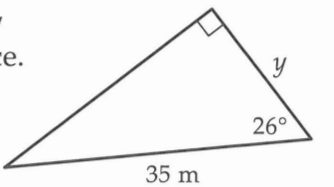
B

The cosine ratio

$$\cos A = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

EXAMPLE:

Find the length of side y correct to 1 decimal place.



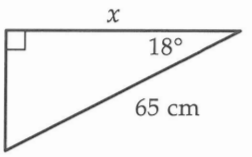
$$\frac{y}{35} = \cos 26°$$

$$y = 35 \cos 26°$$

$$y = 31.5 \text{ m}$$

Calculate the length of side x correct to 1 decimal place.

1



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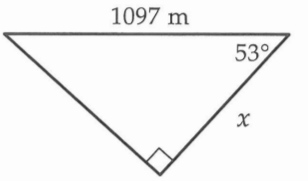

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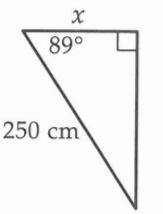

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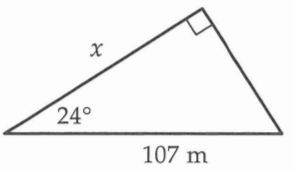

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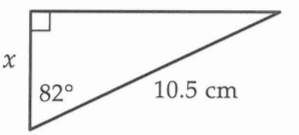

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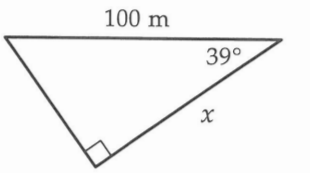

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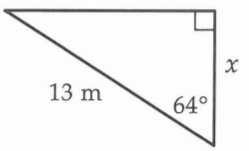

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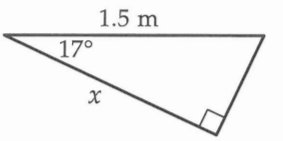

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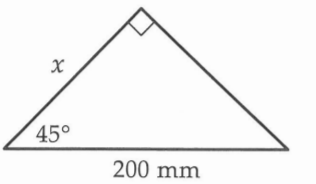

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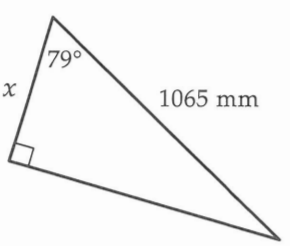

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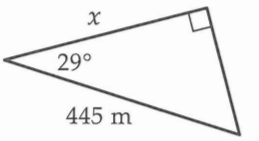

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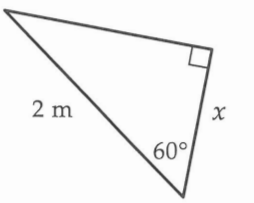

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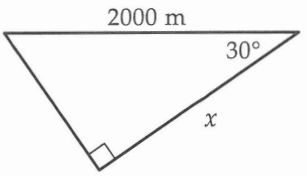
13



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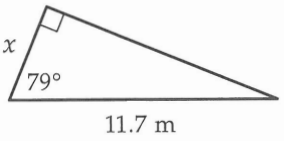

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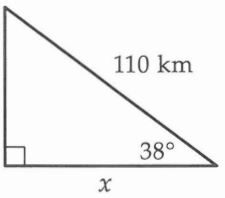

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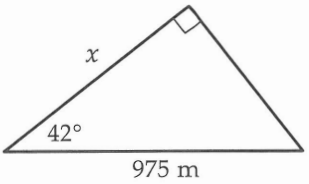

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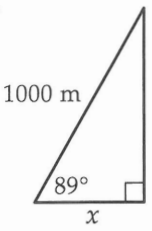

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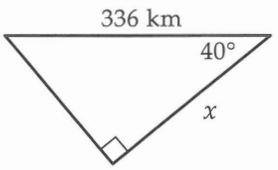
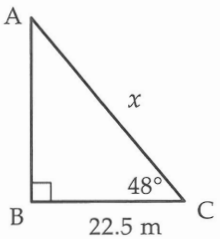


C

EXAMPLE:

Find the length of side x correct to 1 decimal place.



$$\cos 48° = \frac{22.5}{x}$$

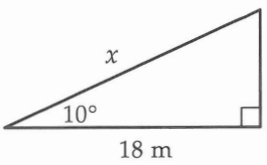
$$x \cos 48° = 22.5$$

$$x = \frac{22.5}{\cos 48°}$$

$$x = 33.6 \text{ m}$$

Use the cosine ratio to calculate the length of side x correct to 1 decimal place.

1



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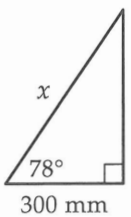

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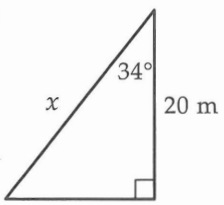

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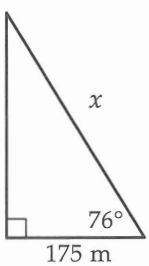

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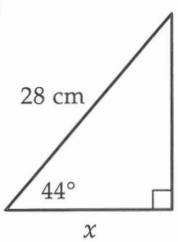

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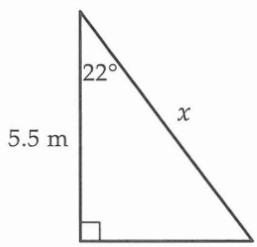

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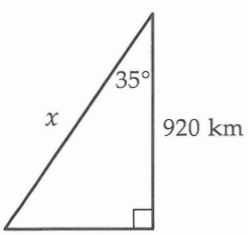

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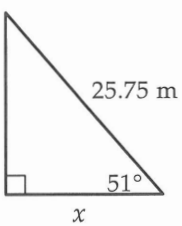

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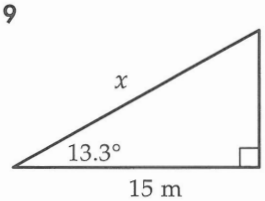

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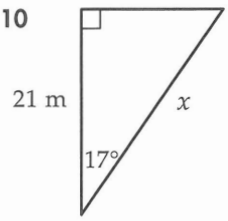

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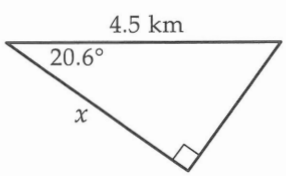

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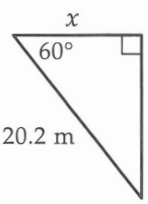

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Note: diagrams not to scale

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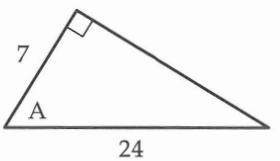
Finding the size of an angle *cosine ratio* 9.1

A Use a calculator to find the size of ∠A to the nearest degree.

1 $\cos A = 0.7071$
2 $\cos A = \frac{4}{5}$
3 $\cos A = 0.5$
4 $\cos A = 0.866$
5 $\cos A = 1$
6 $\cos A = 0$
7 $\cos A = 0.237$
8 $\cos A = \frac{7}{8}$
9 $\cos A = 0.02$
10 $\cos A = \frac{2.5}{6}$
11 $\cos A = 0.5773$
12 $\cos A = \frac{1}{2.5}$

B

EXAMPLE: Find ∠A to the nearest degree.



As the 2 sides given are the adjacent and the hypotenuse, we use the cosine ratio.

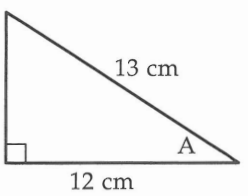
$$\text{cosine ratio} = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\cos A = \frac{7}{24}$$

$$A = 73°$$

Find the size of ∠A to the nearest degree.

1



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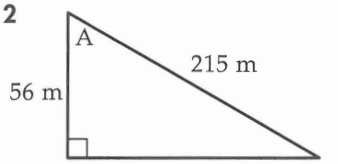

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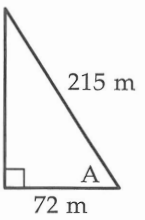

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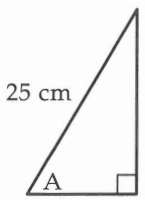

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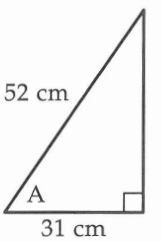

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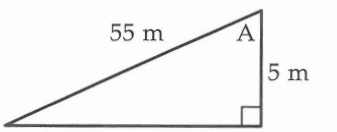

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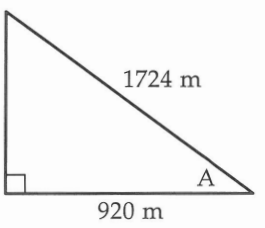

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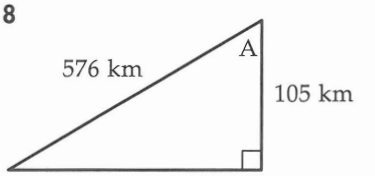

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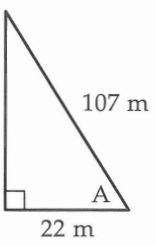

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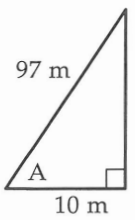

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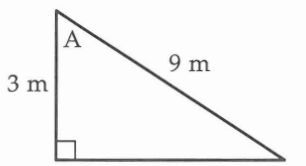

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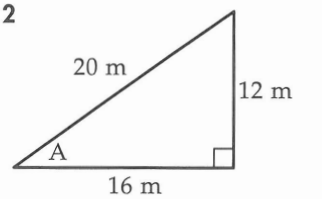

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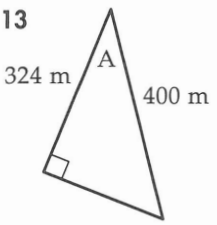

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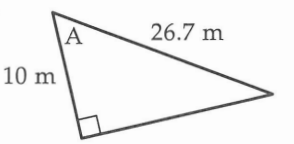

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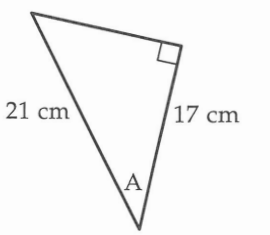

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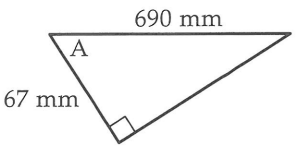
Find the size of ∠A to the nearest degree.

16



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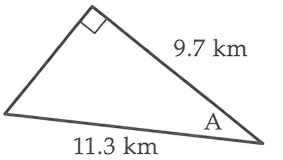

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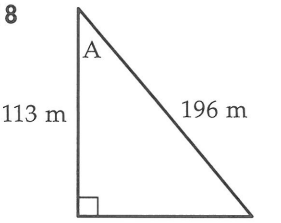

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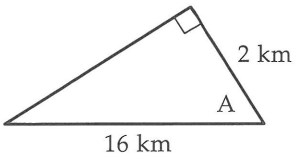

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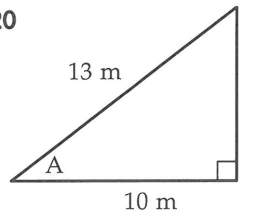

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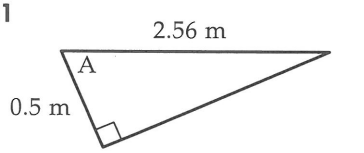

C

For each triangle, calculate the sizes of all 3 angles to 1 decimal place.
(Hint: use the cosine ratio to calculate the size of one unknown angle first.)

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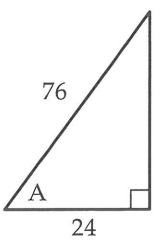

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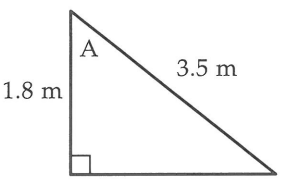

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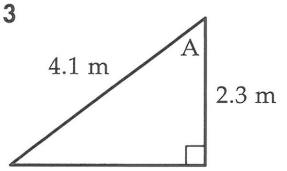

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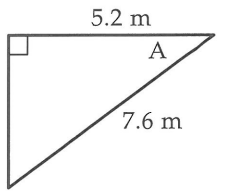

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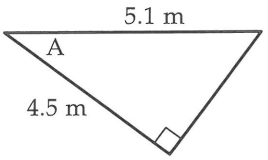

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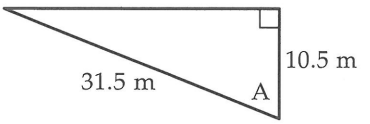

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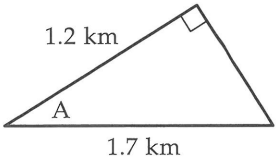

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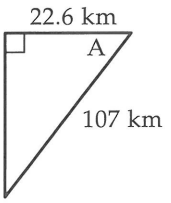

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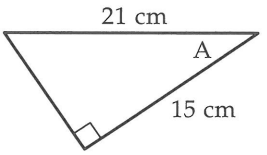

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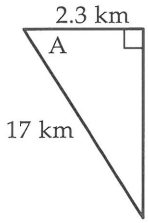

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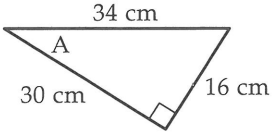

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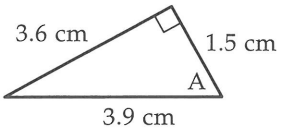

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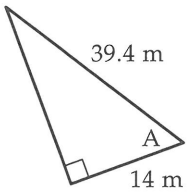

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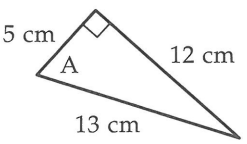

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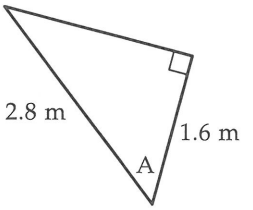

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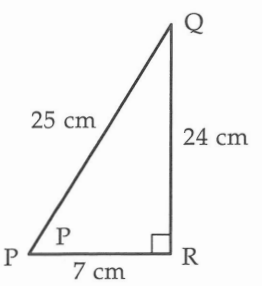
Finding the length of a side *mixed ratios* 10.1

A

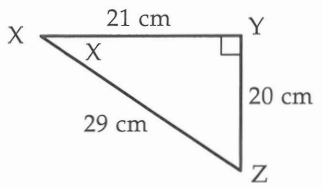


For the triangle PQR, give the length of:

1 the hypotenuse.
2 the side opposite ∠P.
3 the side adjacent to ∠P.

For the triangle PQR, write the value of each ratio as a fraction.

4 tanP 5 sinP 6 cosP



For the triangle XYZ, write the value of each ratio as a fraction.

7 sinX 8 cosX 9 tanX

B Use the tangent, sine or cosine ratio to calculate the length of side x correct to 1 decimal place.

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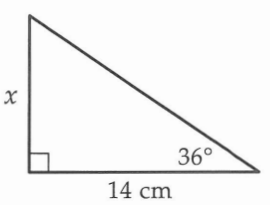

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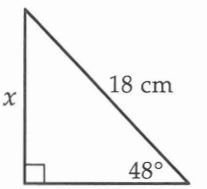

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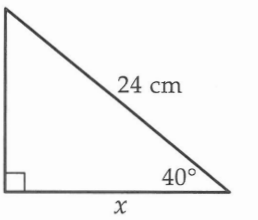

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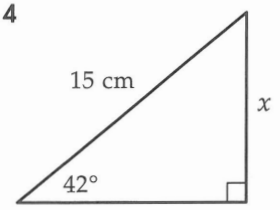

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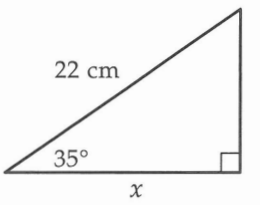

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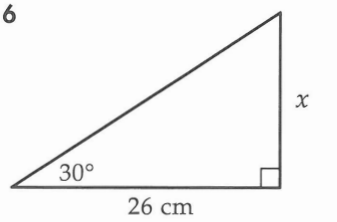

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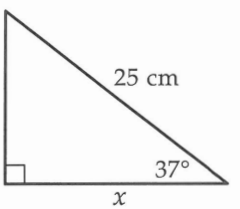

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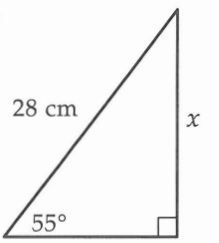

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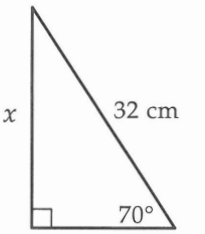

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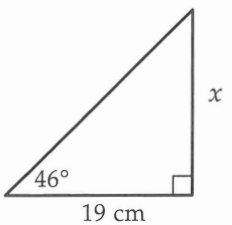

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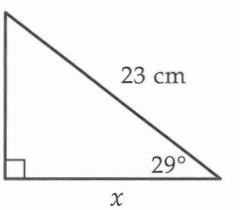

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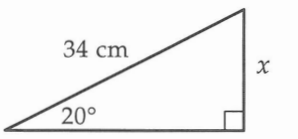

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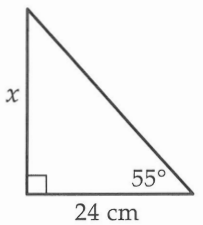

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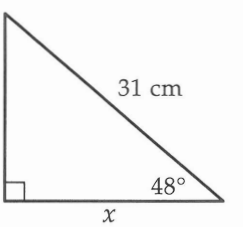

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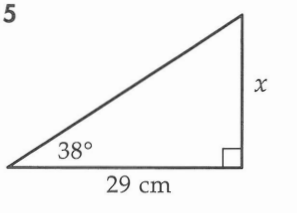

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C Calculate the length of side a correct to 1 decimal place.

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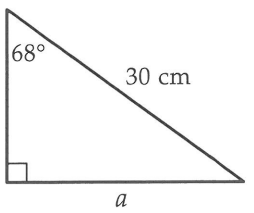

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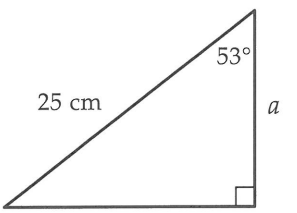

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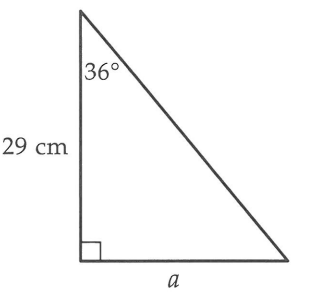

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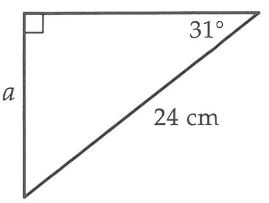

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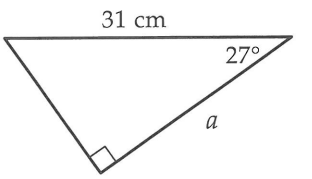

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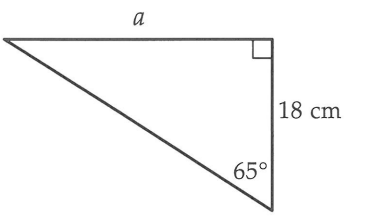

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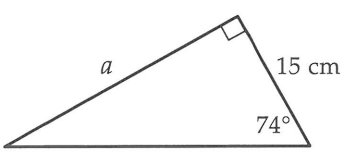

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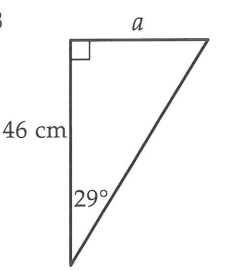

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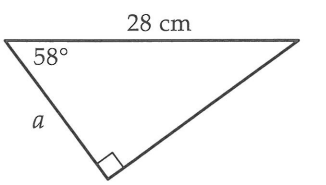

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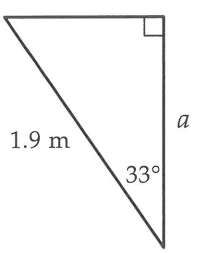

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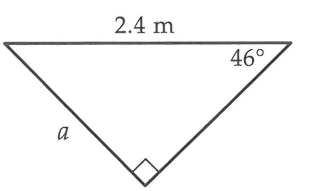

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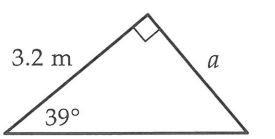

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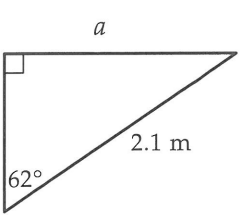

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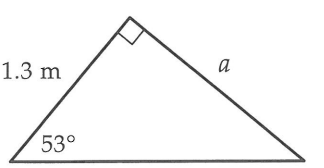

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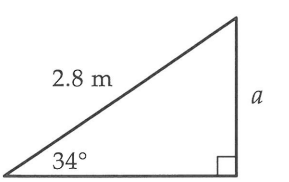

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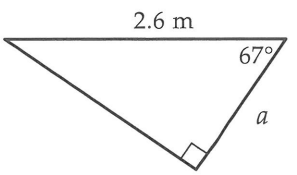

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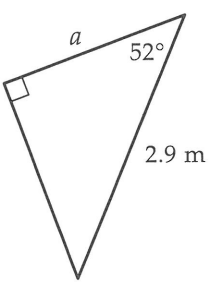

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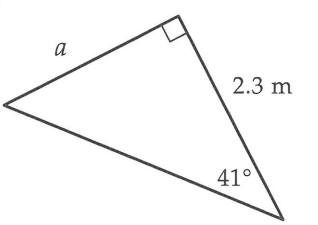

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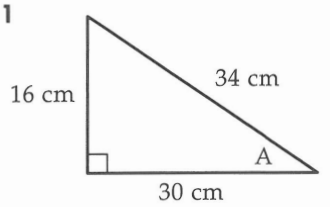
Finding the size of an angle *mixed ratios* 11.1

A Write the values of sin A, cos A and tan A for each triangle. Leave each ratio in simplest fraction form.

1



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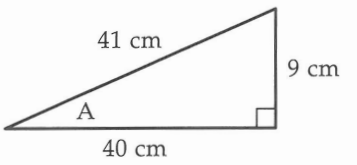

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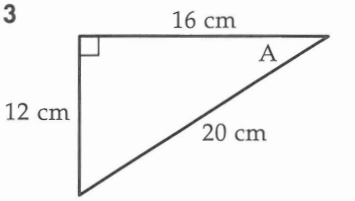
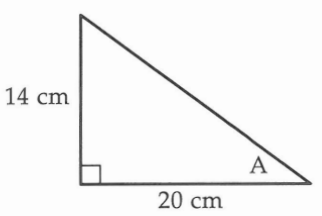


B Use the tangent, sine or cosine ratio to find the size of angle A to the nearest degree.

1



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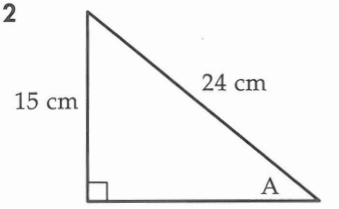

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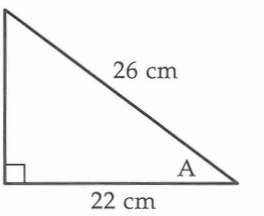

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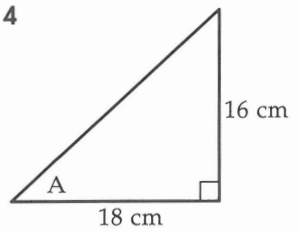

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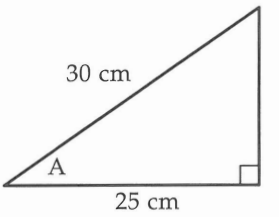

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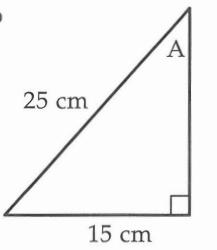

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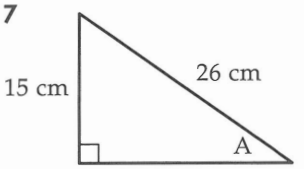

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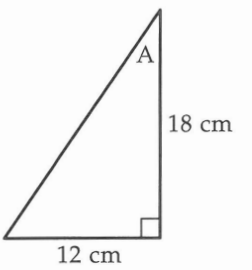

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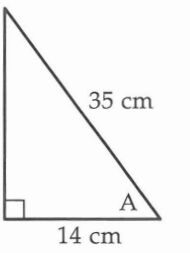

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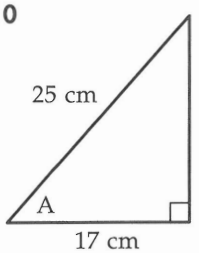

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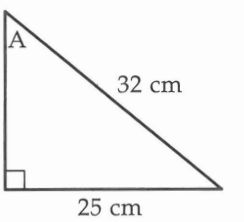

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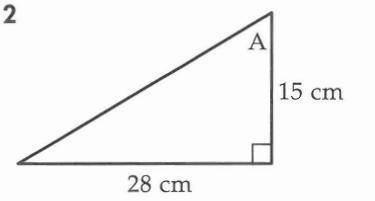

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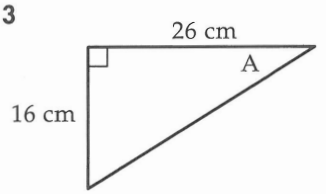

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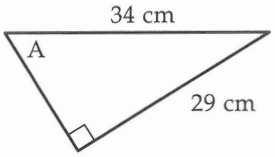

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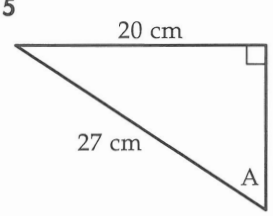

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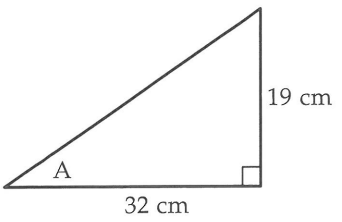
C Calculate the size of angle A to the nearest degree.

1



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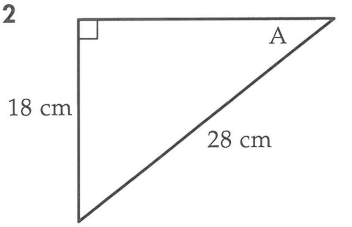

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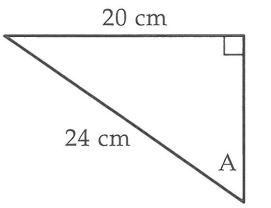

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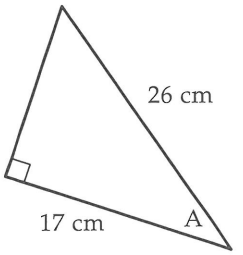

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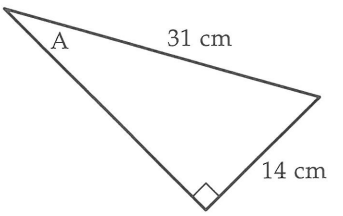

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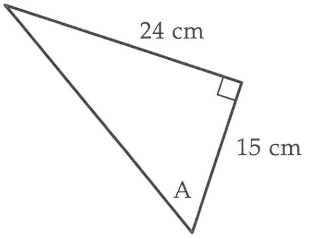

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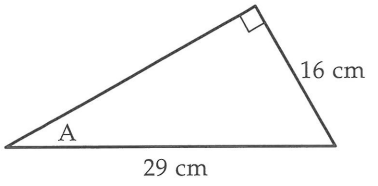

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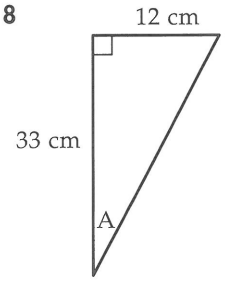

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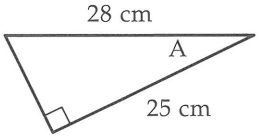

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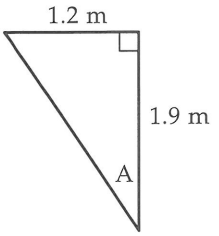

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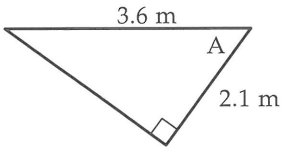

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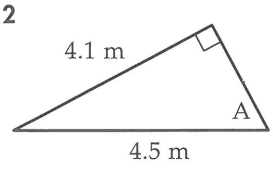

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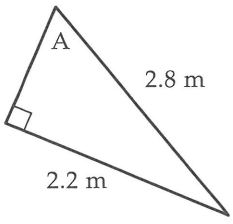

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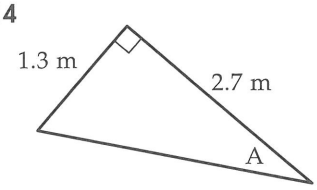

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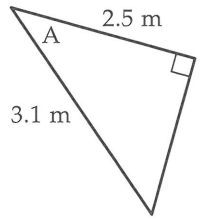

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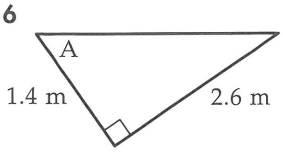

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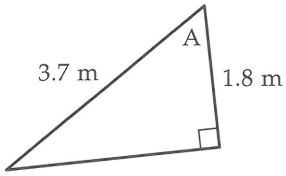

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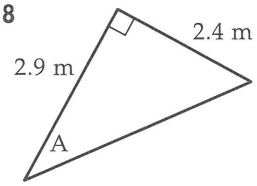

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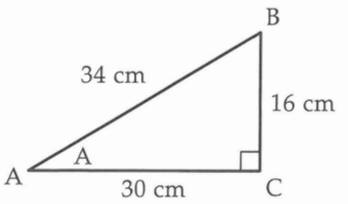
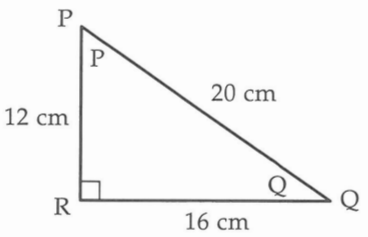
Finding the length of the hypotenuse 12.1

A



For the triangle ABC, write the length of:

1 the hypotenuse.
2 the side opposite ∠A.
3 the side adjacent to ∠A.



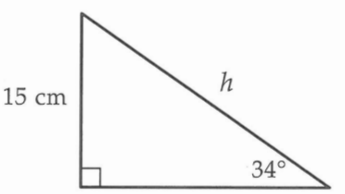
For the triangle PQR, write the length of:

4 the hypotenuse.
5 the side opposite ∠P.
6 the side adjacent to ∠P.
7 the side opposite ∠Q.
8 the side adjacent to ∠Q.

B

EXAMPLE:

Find the length of the hypotenuse correct to 1 decimal place.



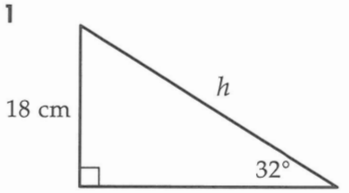
$$\sin 34° = \frac{\text{opposite side}}{\text{hypotenuse}} = \frac{15}{h}$$

rearrange: $h = 15 \div \sin 34°$

$= 26.8$ cm

Use the **sine** ratio to calculate the length of the hypotenuse correct to 1 decimal place.

1



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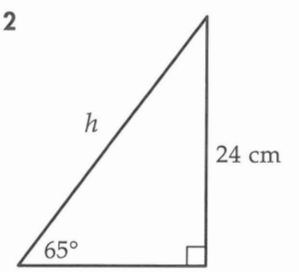

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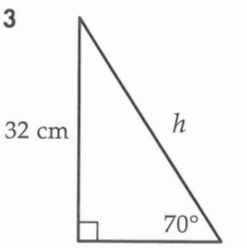

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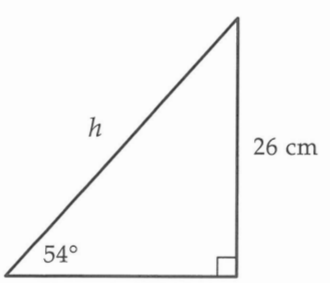

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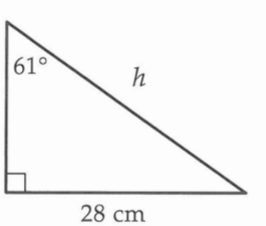

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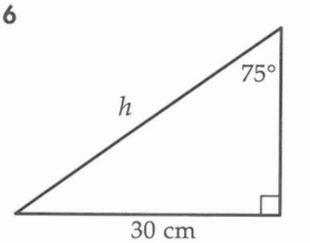

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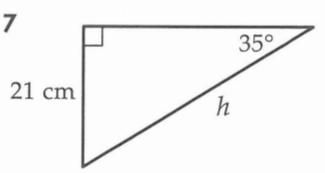

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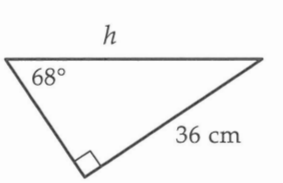

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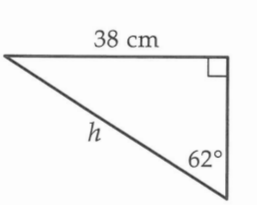

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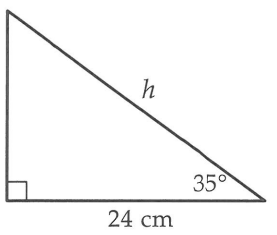
Use the **cosine** ratio to calculate the length of the hypotenuse correct to 1 decimal place.

10



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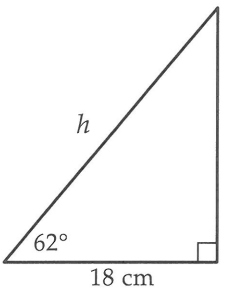

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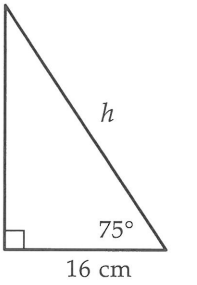

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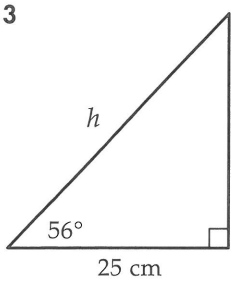

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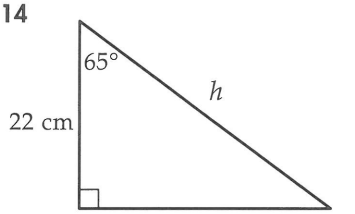

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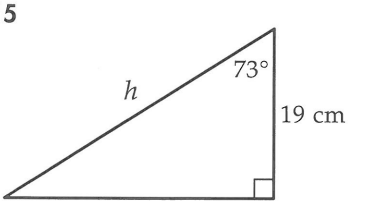

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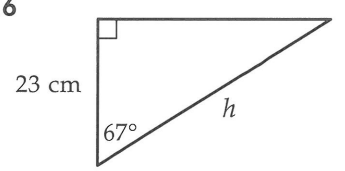

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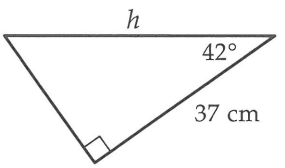

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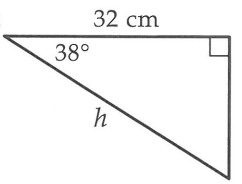
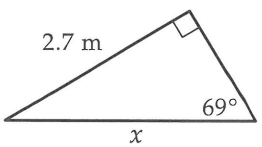


C Calculate the length of the hypotenuse correct to 1 decimal place.

1



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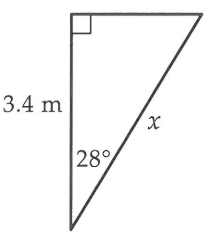

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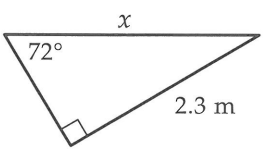

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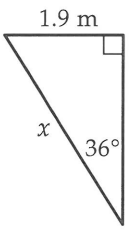

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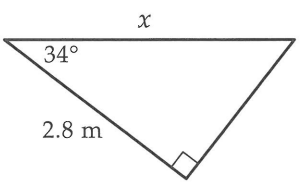

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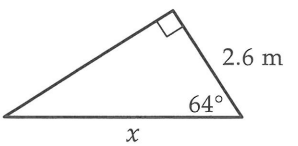

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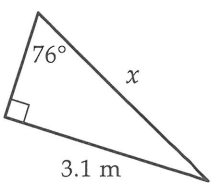

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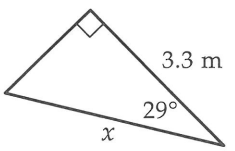

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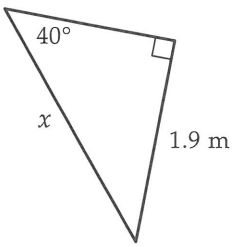

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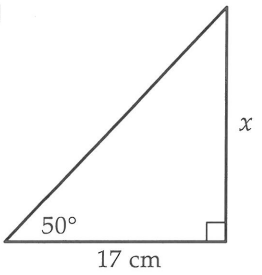
A Calculate the length of side x correct to 1 decimal place.

1



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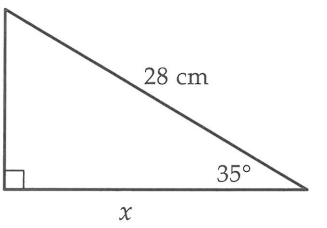

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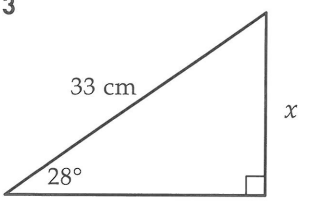

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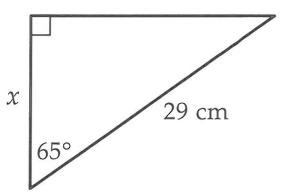

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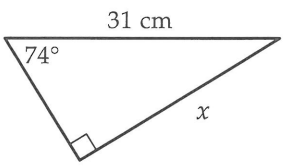

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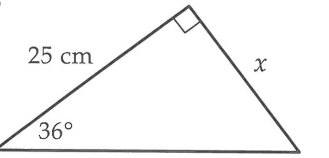
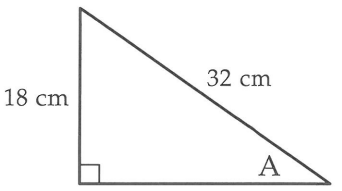


Calculate the size of angle A to the nearest degree.

7



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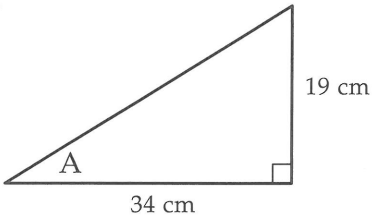

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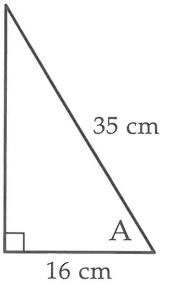

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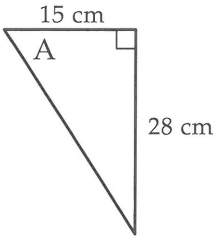

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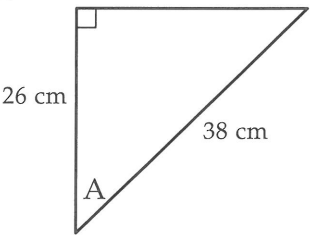

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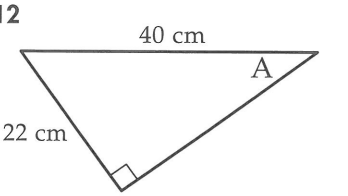
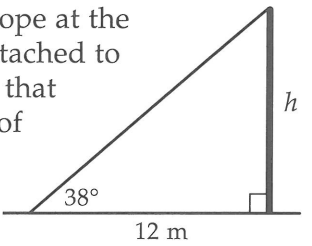
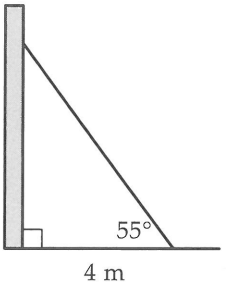


B

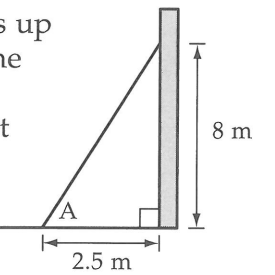
1 A flagpole has a guy rope at the top of the pole. It is attached to a point on the ground that is 12 m from the base of the pole. The rope makes an angle of 38° with the ground. How high is the flagpole?



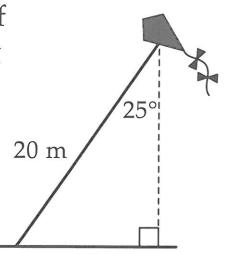
2 A ladder is leaning against a wall, making an angle of 55° with the ground. The foot of the ladder is 4 metres from the wall. How far up the wall does the ladder reach?



3 A ladder reaches 8 metres up a wall when the foot of the ladder is 2.5 metres from the base of the wall. What angle does the ladder make with the ground?



4 A kite is flying on the end of a 20-metre string. The string makes an angle of 25° with the vertical. How high is the kite?



Note: diagrams not to scale

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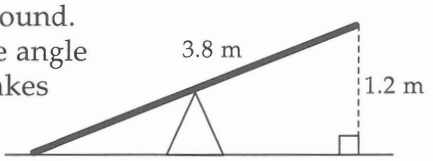
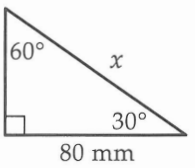
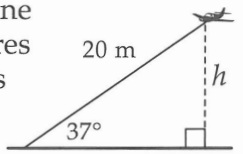
5 The beam of a see-saw is 3.8 metres long. One end of the beam rests on the ground and the other end is 1.2 metres above the ground. Calculate the angle the beam makes with the ground.



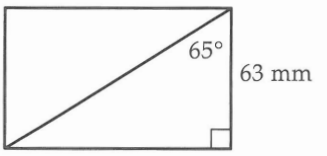
6 On a 30°–60° set square, the side opposite the 60° angle is 80 mm long. Find the length of the longest side to the nearest millimetre.



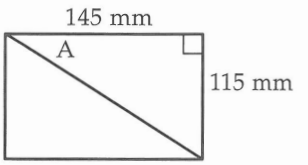
7 A girl is flying a model plane with a control wire 20 metres long. When the wire makes an angle of 37° with the horizontal, how much higher than the girl's hand is the plane?



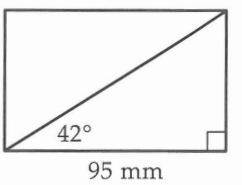
8 A rectangle is 63 mm wide. The diagonal makes an angle of 65° with the width. Calculate the length of the rectangle to the nearest millimetre.



9 A rectangle is 145 mm long and 115 mm wide. What is the size of the angle between the diagonal and the longer side?



10 A rectangle is 95 mm long. The diagonal makes an angle of 42° with the longer side. Calculate the length of the diagonal to the nearest millimetre.



C

Use the information in each question to draw a diagram. Label the diagram with the given information and then answer the question.

1 An 8-metre ladder is leaning against a wall. It makes an angle of 25° with the wall. How far up the wall does the ladder reach?

2 On a 30°–60° set square, the side opposite the 60° angle is 155 mm long. Find the length of the side opposite the 30° angle to the nearest millimetre.

3 A ladder is 6 metres long. It is leaning against a wall, making an angle of 58° with the ground. How far up the wall does the ladder reach? (Answer to 1 decimal place.)

4 A boat at anchor has 50 metres of its anchor chain in the water. The chain makes an angle of 40° with the horizontal. How deep is the water to the nearest metre?

5 A flagpole is supported by a wire stay. The wire is attached 12 metres above the base of the pole and makes an angle of 65° with the ground. How long is the wire? (Answer to 1 decimal place.)

6 The diagonal of a rectangle makes an angle of 56° with the shorter sides. Given that the diagonal is 185 mm long, what is the width of the rectangle to the nearest millimetre?

7 A 5-metre ladder reaches 4.3 metres up a vertical wall. What is the size of the angle between the ladder and the wall?

8 A boy is flying a kite on 100 metres of string. The string makes an angle of 41° with the horizontal. How much higher is the kite than the boy's hand? (Answer to 1 decimal place.)

9 A 3.6-metre ladder leans against a window sill 2.5 metres above the ground. Find the size of the angle between the ladder and the ground.

10 A rectangle is 180 mm long and 126 mm wide. What is the size of the angle between the diagonal and the shorter sides?

11 Find the angles of a rhombus whose diagonals measure 16 cm and 22 cm. (Hint: the diagonals of a rhombus bisect each other at right angles.)

12 The sides of a rhombus measure 240 mm. One of the angles of the rhombus is 50°. Calculate the lengths of the two diagonals to the nearest millimetre.

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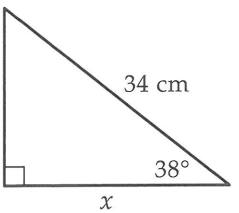
Angles of elevation and depression 14.1

A Calculate the length of side x correct to 1 decimal place.

1



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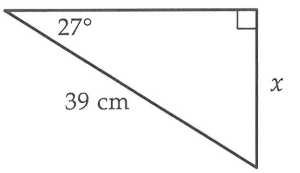

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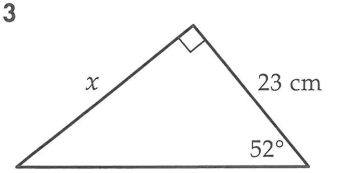
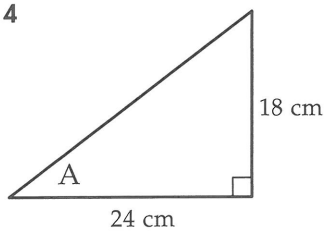


Calculate the size of angle A to the nearest degree.

4



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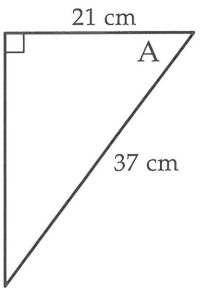

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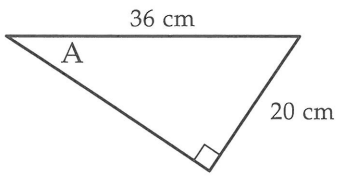
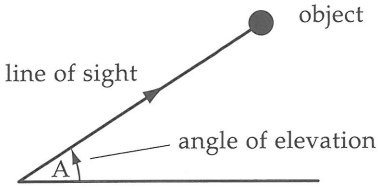


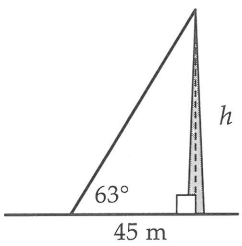
B

When you are **looking up** towards an object, the **angle of elevation** is the angle between your line of sight and the horizontal.



EXAMPLE:

A girl is standing 45 metres from the base of a tower. She measures the angle of elevation to the top of the tower as 63°. Calculate the height of the tower correct to 1 decimal place.

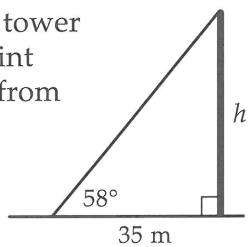


$$\frac{h}{45} = \frac{\text{opposite side}}{\text{adjacent side}} = \tan 63°$$

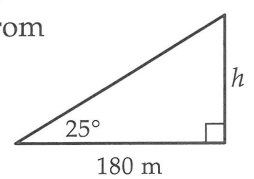
$$h = 45 \tan 63°$$

$$= 88.3 \text{ m}$$

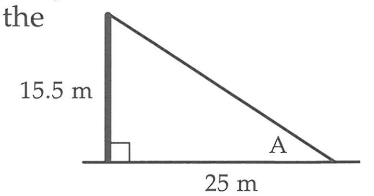
1 The angle of elevation of a tower is 58°, measured from a point 35 metres on level ground from the tower's base. Calculate the height of the tower to the nearest metre.



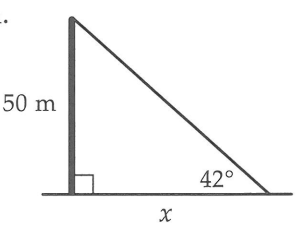
2 The angle of elevation of a vertical cliff is measured from a boat as 25°. The boat is 180 metres from the base of the cliff. How high is the cliff to the nearest metre?



3 A vertical pole is 15.5 metres high. It casts a shadow 25 metres long on horizontal ground. Calculate the angle of elevation of the sun to the nearest degree.



4 A boy lying on level ground measured the angle of elevation of a tower as 42°. The tower is 50 metres high. How far is the boy from the base of the tower? (Answer to the nearest metre.)



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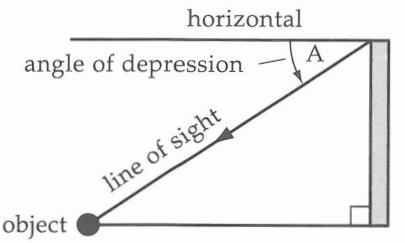
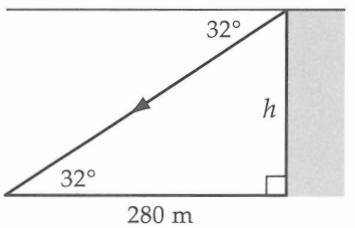
When you are **looking down** towards an object, the **angle of depression** is the angle between your line of sight and the horizontal.

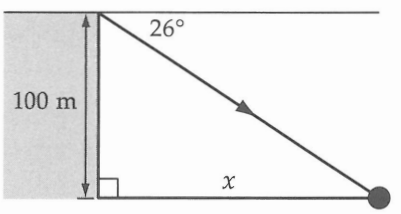
Remember, the two horizontal lines are parallel.



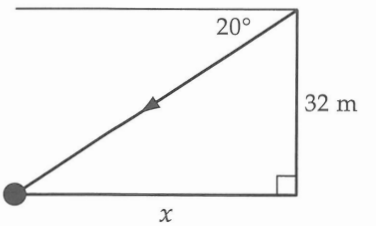
5 A boat is 280 m from the base of a cliff. The angle of depression of the boat from the top of the cliff is 32°. Calculate the height of the cliff to the nearest metre. (Hint: Use the angle properties of parallel lines to identify the equal angle inside the triangle.)



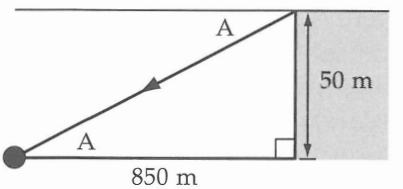
6 A boy standing on top of a 100-metre high cliff can see a boat at sea. He measures the angle of depression of the boat as 26°. How far is the boat from the foot of the cliff? (Answer to the nearest metre.)



7 The angle of depression of an object on level ground is measured as 20° from the top of a tower 32 metres high. How far is the object from the base of the tower, correct to 1 decimal place?



8 A boat is 850 m from the base of a 50-metre cliff. Calculate the angle of depression of the boat from the top of the cliff to the nearest degree.



C

Use the information in each question to draw a diagram. Label the diagram with the given information and then answer the question.

1 A tree casts a shadow 24 metres long on level ground. The angle of elevation of the sun is 35°. Calculate the height of the tree correct to 1 decimal place.

2 A tree is 28 metres tall and casts a shadow 32 metres long. Calculate the angle of elevation of the sun to the nearest degree.

3 From a boat at sea, a lighthouse 75 metres above sea level is observed at an angle of elevation of 6°. How far is the boat from the base of the lighthouse? (Answer to the nearest metre.)

4 A boat is 600 metres from the base of a vertical cliff. From the top of the cliff, the angle of depression of the boat is 12°. Calculate the height of the cliff correct to 1 decimal place.

5 From the top of a 165-metre building, the angle of depression of a car is 42°. How far is the car from the building along level ground? (Answer to the nearest metre.)

6 At a theatre a woman is sitting 20 m from the stage by line of sight. She is also 4 m above the level of the stage. At what angle of depression does she see the stage?

7 From a lighthouse 110 metres above sea level a ship is observed 1.3 km out to sea. What is the angle of depression from the lighthouse to the ship to the nearest degree?

8 A weather balloon is 1.5 km above the ground, at an angle of elevation of 34° from an observer. What is the distance along the line of sight from the balloon to the observer? Answer to the nearest metre.

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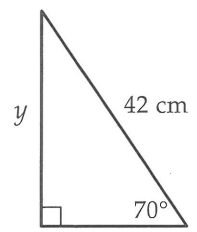
Further problems *right-angled triangles* 15.1

A Calculate the length of side y correct to 1 decimal place.

1



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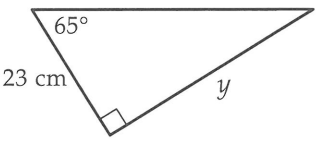

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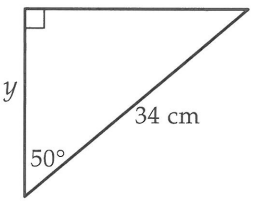
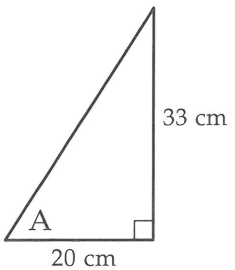


Calculate the size of angle A to the nearest degree.

4



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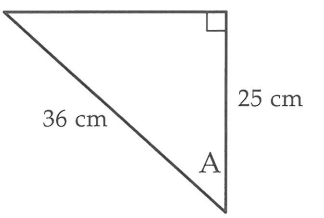

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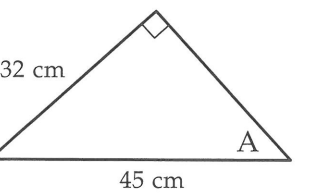
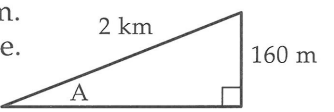
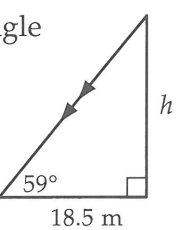


B

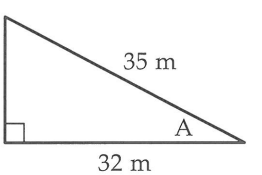
1 A road rises 160 metres in a straight length of 2 km. Find the angle of slope.



2 When the sun's rays are at an angle of 59° to the horizontal, a tree casts a shadow 18.5 metres long. Calculate the height of the tree to 1 decimal place.

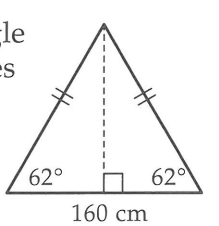


3 An escalator ramp in a shopping centre has a length of 35 metres and covers a horizontal distance of 32 metres. What is the angle of inclination, to the nearest degree?



4 The vertical distance between the top and bottom of a ramp is 1.6 metres. The ramp makes an angle of 10° with the horizontal ground. Calculate the length of the ramp correct to 1 decimal place.

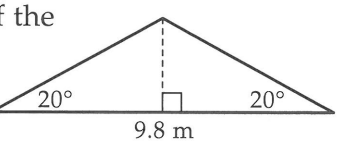
5 The base of an isosceles triangle is 160 cm long. The base angles each measure 62°. Calculate the perpendicular height of the triangle to the nearest centimetre.



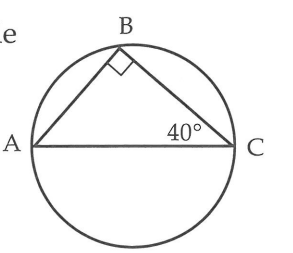
6 The perpendicular height of an isosceles triangle is 85 cm and each base angle is 39°. Calculate the length of the base to the nearest centimetre.

7 The base of an isosceles triangle is 95 cm long and its perpendicular height is 120 cm. Calculate the size of the base angles to the nearest degree.

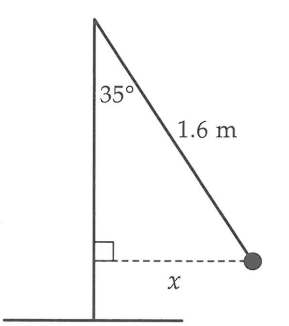
8 The diagram shows a pitched roof with a span of 9.8 m and a pitch of 20°. Calculate the vertical height of the roof to the nearest millimetre.



9 AC is a diameter of a circle with radius 85 mm. ∠ABC = 90° and ∠BCA = 40°. Calculate the length of the chord AB to the nearest millimetre.



10 Ken is on a swing that is 1.6 m long. The swing is pulled back so that it makes an angle of 35° with the vertical frame. How far from the frame is Ken? (Answer correct to 1 decimal place.)



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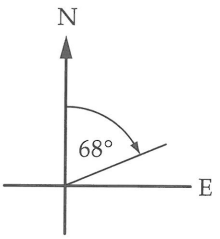
C

Bearings use angles to show the direction of one point from another point.

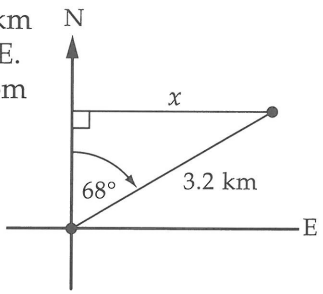
Compass bearings use the four directions of the compass (north, south, east and west) and an angle between 0° and 90°.

EXAMPLE:

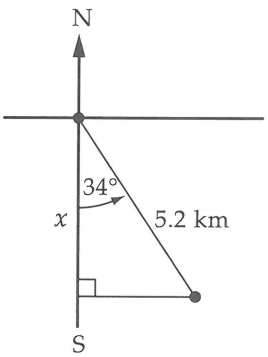
Compass bearing N68°E: Start facing due north and turn through an angle of 68° towards east.



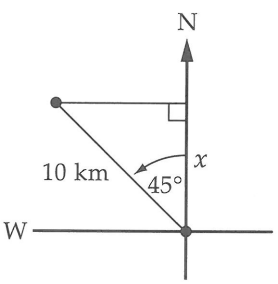
1 Peter walked for 3.2 km on a bearing of N68°E. How far east is he from his starting point? Answer correct to 1 decimal place.



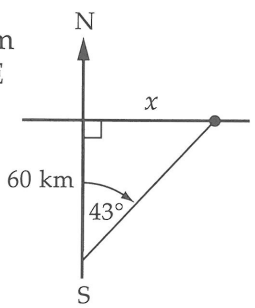
2 A bushwalker left her base camp walking on a bearing of S34°E. She walked 5.2 km. How far south of her base camp is she? Answer correct to 1 decimal place.



3 A small plane takes off and flies NW for 10 km, then turns and flies due east until it is due north of the airstrip. How far from the airstrip is the plane? Answer correct to 1 decimal place.



4 A ship sails south for 60 km then turns and sails N43°E until it is due east of its starting point. How far is the ship from its starting point? Answer correct to 1 decimal place.



5 A ship sails due north, then turns and sails 80 km due west. The ship is now on a bearing of N65°W from its starting point. How far due north did the ship sail? Answer to the nearest kilometre.

6 After walking due east and then turning and walking due south, a bushwalker is 5.5 km, S35°E from his starting point. How far due south did he walk? Answer correct to 1 decimal place.

7 A ship sails 145 km on a bearing of S58°W. How far south of its starting point is the ship? Answer to the nearest kilometre.

8 Two cars leave from the same starting point. one car travels due west, the second car travels N25°W. After travelling 20 km, the second car is due north from the first car. How far apart are the cars? Answer correct to 1 decimal place.

9 A ship sails from a port on a course bearing S32°E for 28 nautical miles. How far east of the port is the ship?

10 A small plane takes off from an airstrip and flies on a course S55°W for 26 km. It then turns and flies due north, until it is due west of the airstrip. How far has the plane travelled?

11 An observer on one bank of a straight section of a river observes a tree on the opposite bank, on a bearing of N63°W. On moving 150 metres downstream, the bearing of the same tree is N63°E. How wide is the river?

12 Two ships sail from a port. One sets a course of N58°E, the other sets a course of S58°E. What is the distance between the two ships when both have travelled 40 km from the port?

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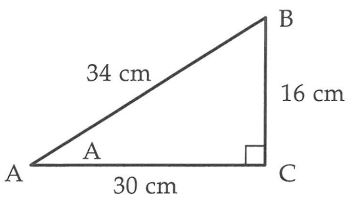
Review A: Trigonometry *right-angled triangles* 16.1

A



For the triangle ABC, write the length of:

1 the hypotenuse.
2 the side opposite ∠A.
3 the side adjacent to ∠A.

For the triangle ABC, write each ratio as a fraction in simplest form.

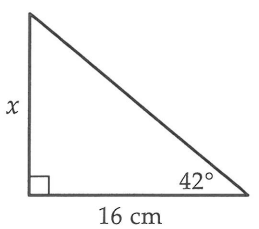
4 $\sin A$ 5 $\tan A$ 6 $\cos A$

Use a calculator to find each ratio correct to 3 decimal places.

7 $\sin 56°$ 8 $\tan 28°$ 9 $\cos 35°$ 10 $\tan 72°$

B Calculate the length of side x correct to 1 decimal place.

1



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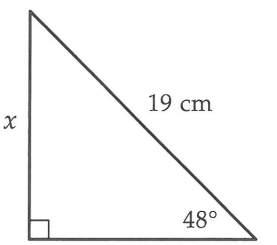

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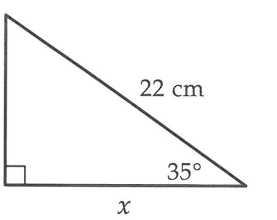

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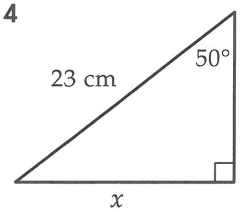

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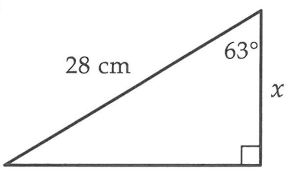

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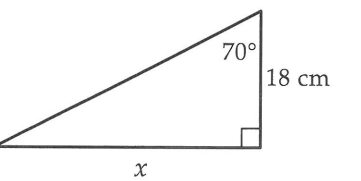

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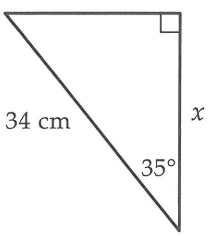

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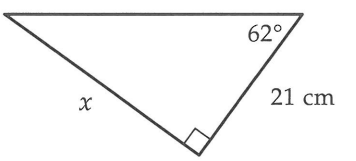

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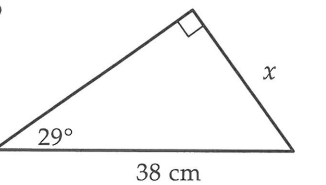

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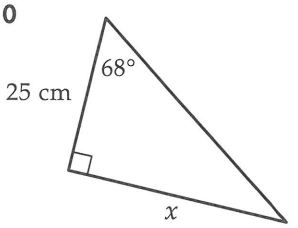

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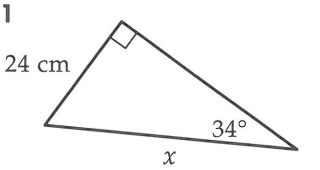

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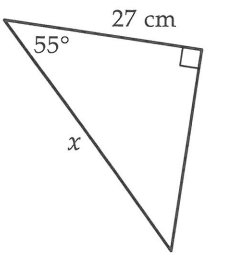
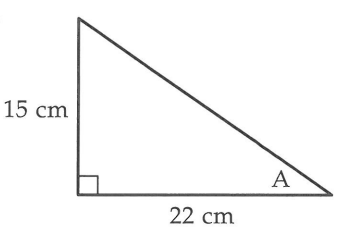


Find the size of angle A to the nearest degree.

13



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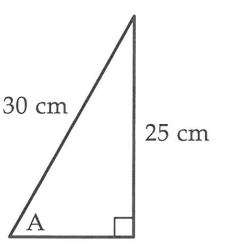

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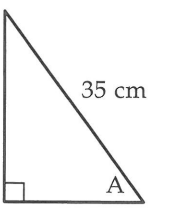

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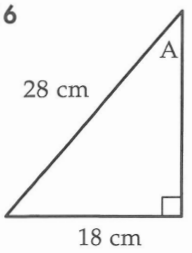
Find the size of angle A to the nearest degree.

16



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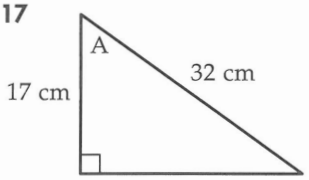

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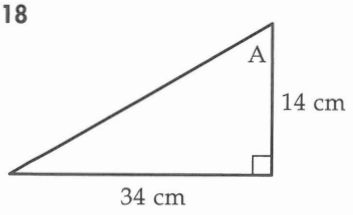

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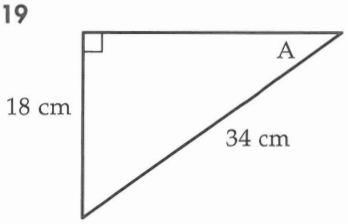

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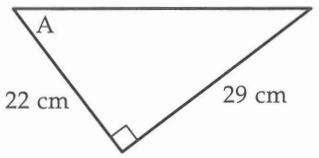

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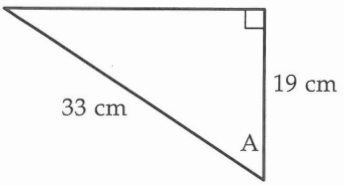

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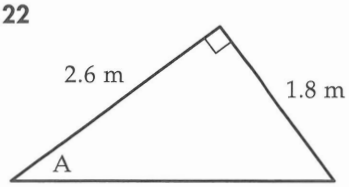

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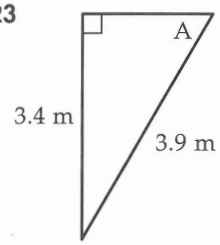

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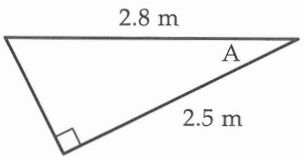

C

1. A ladder leans against a wall, making an angle of 65° with the ground. The ladder is 3.2 metres long. How far up the wall does the ladder reach? (Answer to 1 decimal place.)
2. A kite is flying on the end of a 24-metre string. The string makes an angle of 32° with the vertical. How high is the kite? (Answer to 1 decimal place.)
3. A rectangle is 95 mm long. The diagonal makes an angle of 42° with the longer sides. Calculate the width of the rectangle to the nearest millimetre.
4. A flagpole is supported by a wire stay. The wire is attached 14 metres above the base of the pole and makes an angle of 63° with the ground. How long is the wire? (Answer to 1 decimal place.)
5. A ladder 4.2 metres long reaches 3.6 metres up a vertical wall. What is the size of the angle between the ladder and the ground?
6. A boy is standing 50 metres from the base of a tower. He measures the angle of elevation of the top of the tower as 59°. Calculate the height of the tower, correct to 1 decimal place.
7. The angle of depression of an object on level ground is measured as 25° from the top of a building 34 metres high. How far is the object from the base of the building? (Answer to the nearest metre.)
8. The vertical distance between the top and bottom of a ramp is 2.3 metres. The ramp makes an angle of 12° with the horizontal ground. Calculate the length of the ramp, correct to 1 decimal place.
9. A balloon at a height of 45 metres was observed at an angle of elevation of 33°. How far from the point on the ground directly below the balloon, was the observer?
10. A 120-centimetre pendulum swings 18° on each side of the vertical. How high above its lowest point does the end of the pendulum rise?

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Note: diagrams not to scale

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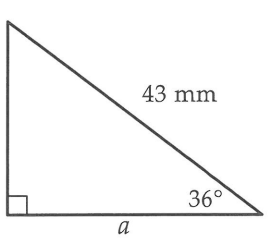
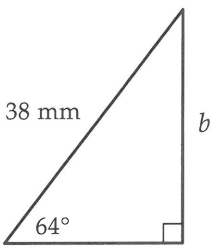
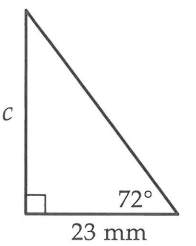
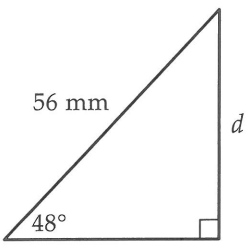
Riddle: Why did the man put his money in the freezer?

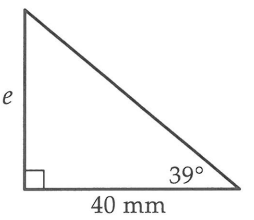
For each triangle, calculate the length of the side marked with the pronumeral to the nearest millimetre. Then write the pronumeral above each correct answer.

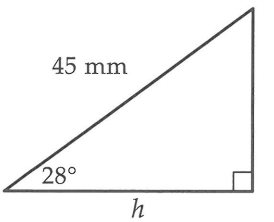


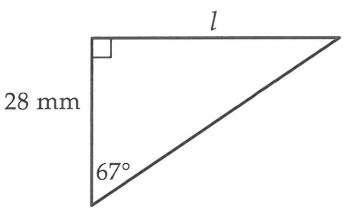


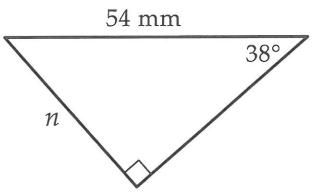


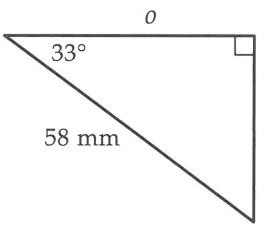


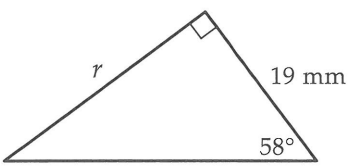


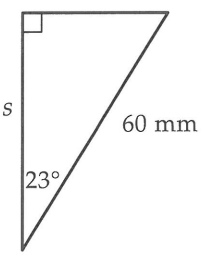


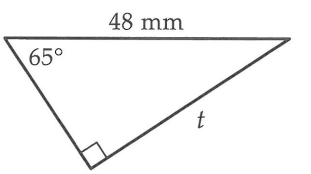


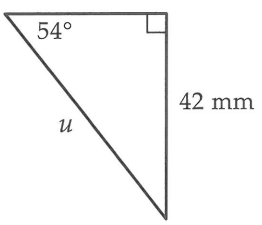


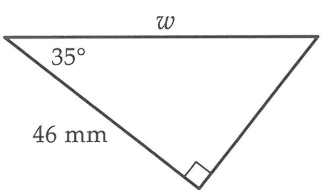












34	32	71	35	52	55	32		40	32		56	35	33	44	32	42		71	49	66	42		40	35	30	42		71	35	55	40

Note: all answers in mm

Note: diagrams not to scale

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A Evaluate each trigonometrical ratio correct to 3 decimal places.

1 tan10°	**2** sin76°	**3** cos55°
4 sin60°	**5** tan45°	**6** sin23.5°
7 cos89°	**8** sin1°	**9** tan89.975°
10 sin45°	**11** cos45°	**12** cos0.01°

B

Make sure that your calculator is operating in degrees.
Remember:

1 degree = 60 minutes
1° = 60′
and 1 minute = 60 seconds
1′ = 60″

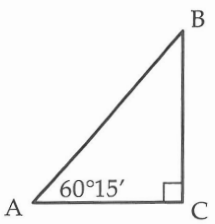
Use the degrees/minutes/seconds (° ′ ″) key on your calculator to enter angles given in degrees and minutes, or degrees, minutes and seconds.

Find each trigonometrical ratio correct to 3 decimal places.

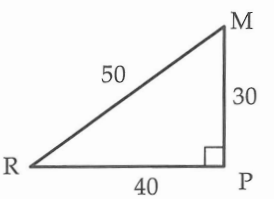
1 tan47°2′	**2** cos46°11′	**3** 5tan11°15′	**4** cos36°36′
5 tan89°59′	**6** sin88°45′	**7** sin23°24′	**8** tan1°11′
9 cos29°59′	**10** cos10°1′	**11** sin55°29′	**12** cos30°10′
13 sin29°59′	**14** cos17°17′	**15** tan22°30′	**16** tan7°5′
17 tan64°38′	**18** cos10°57′	**19** tan31°39′	**20** cos43°25′
21 sin79°3′	**22** sin19°54′	**23** sin7°17′	**24** sin25°25′
25 cos88°55′	**26** cos83°47′	**27** tan23°13′	**28** cos1°5′
29 tan5°1′	**30** cos45°15′	**31** sin28°6′	**32** sin12°56′
33 tan35°8′	**34** tan60°51′	**35** sin52°36′	**36** tan45°1′
37 sin22°2′	**38** cos68°57′	**39** sin44°45′	**40** cos37°12′
41 tan72°16′46″	**42** cos59°59′59″	**43** tan12°10′30″	**44** sin44°56′4″

C **Riddle: What did the lion say to the sea when he got to the beach?**

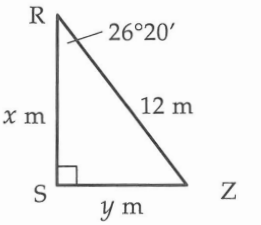
Calculate angles in degrees and lengths and ratios correct to 2 decimal places.



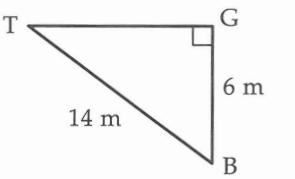
T Find tanA.
R Find ∠B.
O Find cosA.



U Find sinR.
E Find cosR.
L Find tanM.



S Find x.
H Find y.
G Find ∠Z.



O Find ∠T.
A Find ∠B.

1.33	0.8	1.75		0.6	10.75		29°45′	25°23′	64°37′	29°45′	

1.75	0.50	63°40′	0.8	1.75	5.32	0.8	29°45′	

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Note: diagrams not to scale

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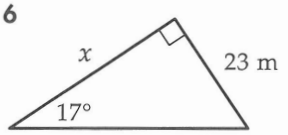
Finding lengths of sides *angles in degrees & minutes* 18.1

A Find the value of x correct to 1 decimal place.

1 $\tan 22°30' = x$ **2** $\sin 11°15' = x$ **3** $\cos 79°59' = x$ **4** $\sin 29°30' = x$

5 $\tan 44°52' = x$

6



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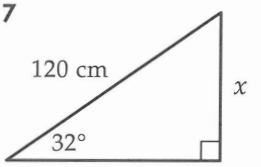

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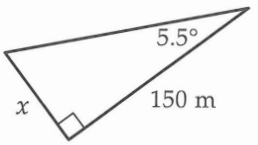

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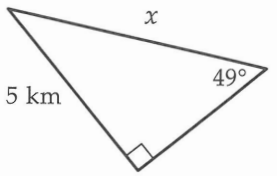
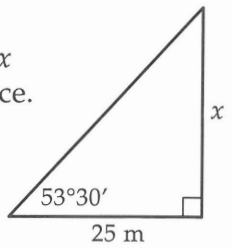


B

Remember: $\tan A = \frac{\text{opp}}{\text{adj}}$ $\sin A = \frac{\text{opp}}{\text{hyp}}$ $\cos A = \frac{\text{adj}}{\text{hyp}}$

EXAMPLE:

Find the length of side x correct to 1 decimal place.



$\frac{x}{25} = \tan 53°30'$

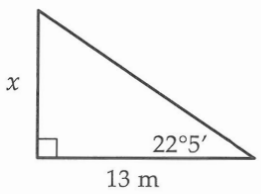
$x = 25 \tan 53°30'$

$= 33.78556095\ldots$

$= 33.8$ m

Calculate the length of side x correct to 1 decimal place.

1



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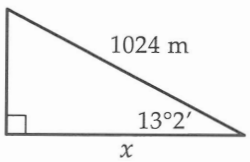

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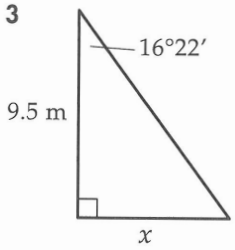

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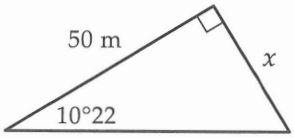

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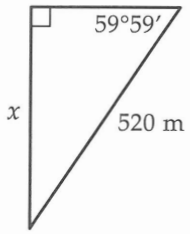

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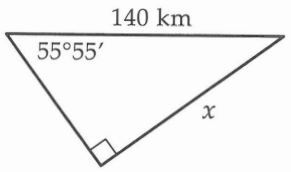

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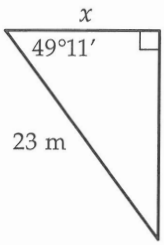

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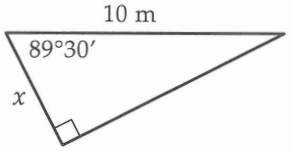

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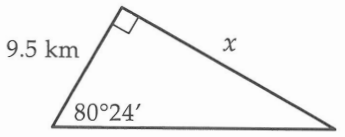

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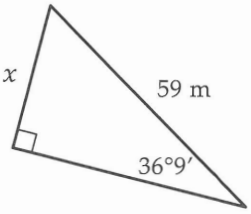

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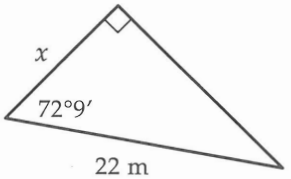

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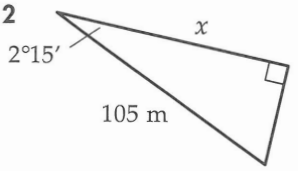

Note: diagrams not to scale

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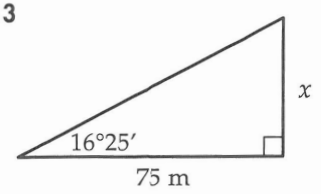
Calculate the length of side x correct to 1 decimal place.

13



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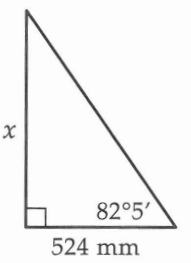
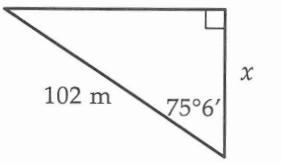


15



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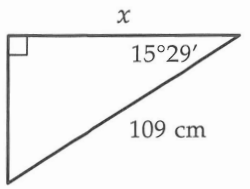

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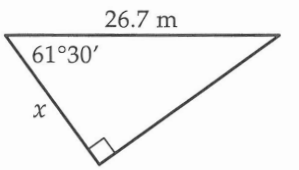

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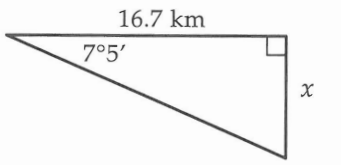
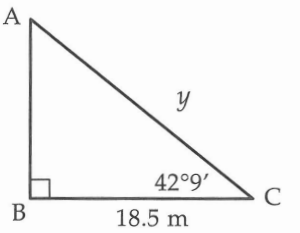


C

EXAMPLE:

Find the length of y correct to 2 decimal places.



$$\cos 42°9' = \frac{18.5}{y}$$

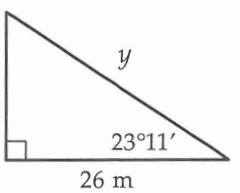
$$y = \frac{18.5}{\cos 42°9'}$$

$$y = 24.95311172\ldots$$

$$= 24.95 \text{ m}$$

Calculate the length of side y correct to 2 decimal places.

1



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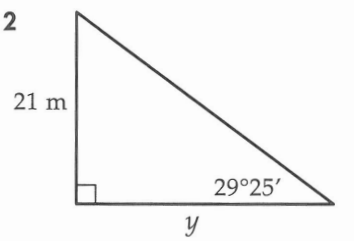

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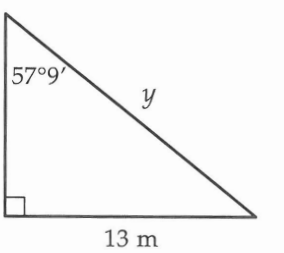

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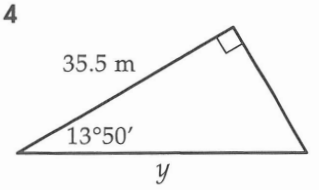

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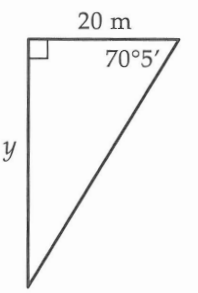

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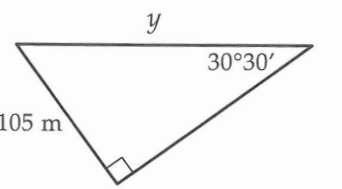

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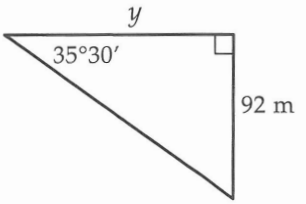

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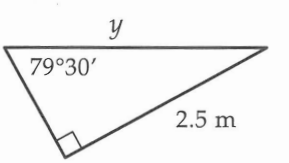

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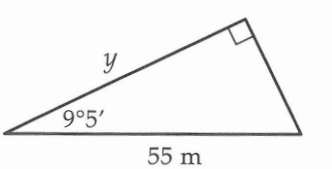

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Note: diagrams not to scale

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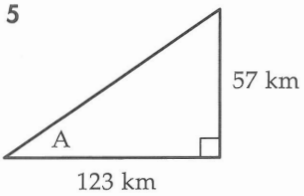
Finding sizes of angles *angles in degrees & minutes* 19.1

A Use your calculator to find the size of ∠A to the nearest degree.

1 $\tan A = \frac{4}{5}$ **2** $\sin A = 0.707$ **3** $\cos A = \frac{2}{3}$ **4** $\sin A = \frac{20}{41}$

5



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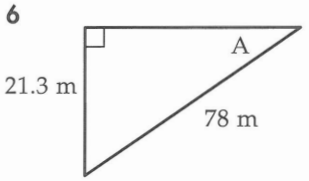

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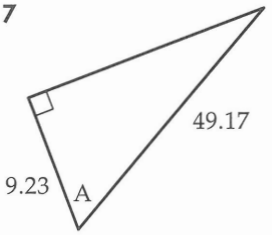

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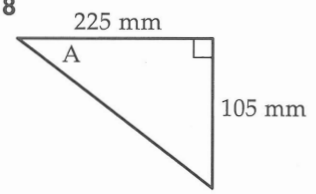
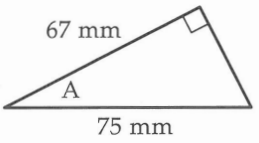


B

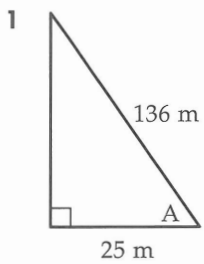
EXAMPLE:
Find ∠A to the nearest minute.



$$\cos A = \frac{67}{75}$$
$$\angle A = 26°42'17.47''$$
$$= 26°42'$$

Calculate the size of ∠A to the nearest minute.

1



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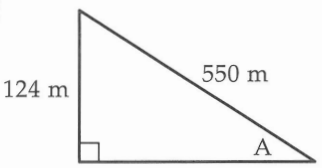

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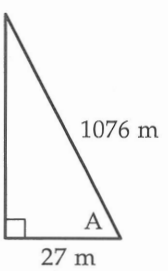

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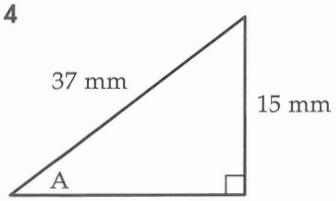

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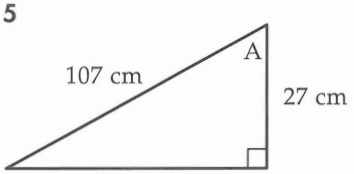

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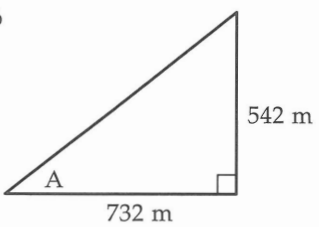

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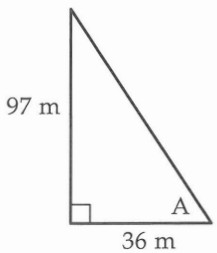

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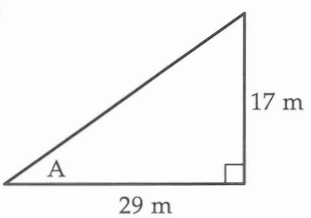

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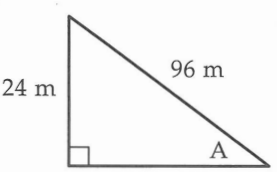

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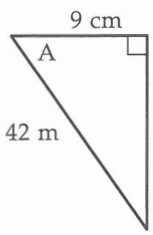

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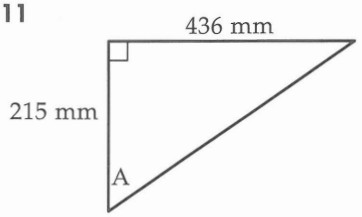

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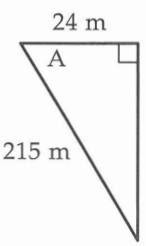

Note: diagrams not to scale
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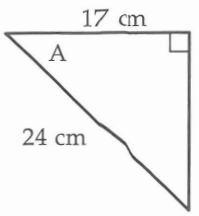
Calculate the size of ∠A to the nearest minute.

13



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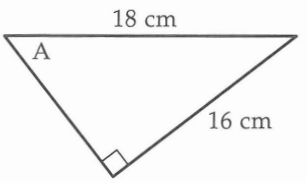

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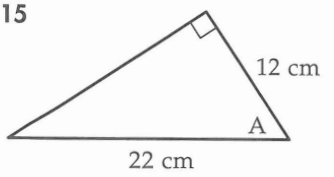

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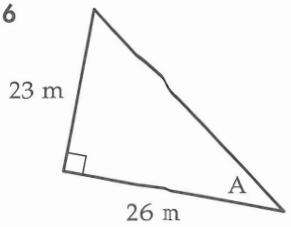

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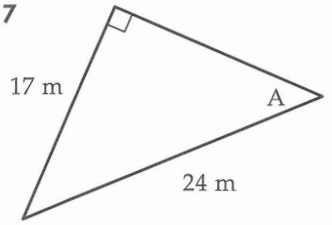

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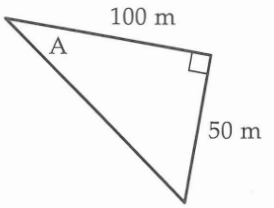
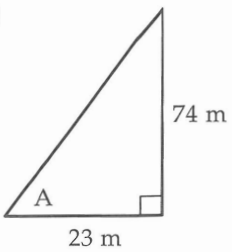


C Calculate the size of ∠A to the nearest minute.

1



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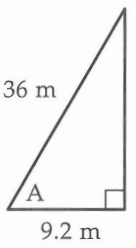

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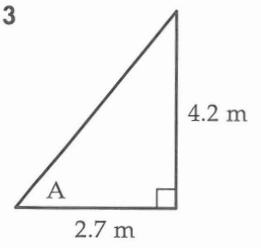

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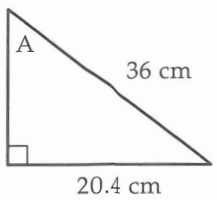

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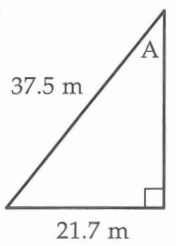

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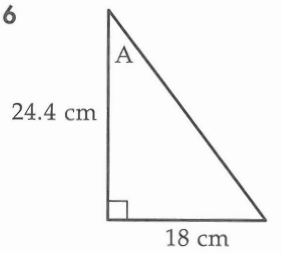

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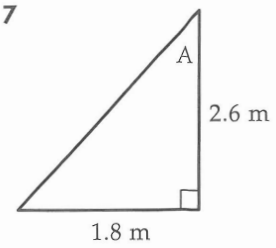

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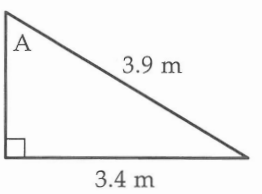

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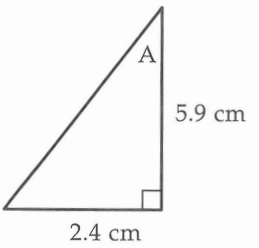

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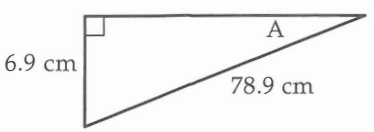

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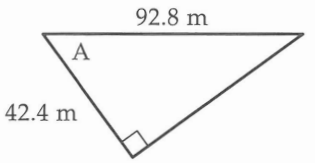

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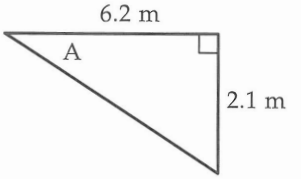

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Note: diagrams not to scale

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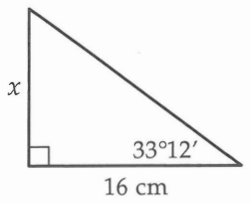
Mixed problems *angles in degrees & minutes* 20.1

A Find the length of side x correct to 1 decimal place.

1



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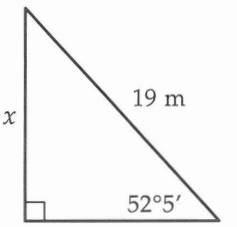

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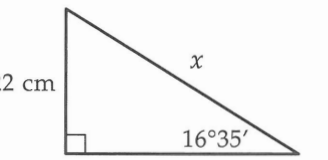

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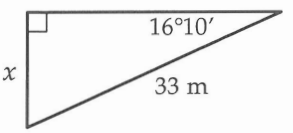

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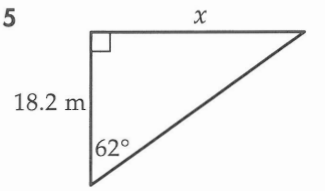

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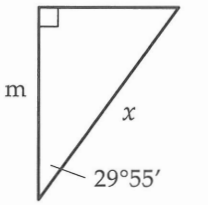

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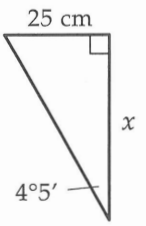

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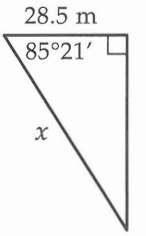

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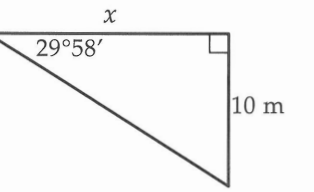
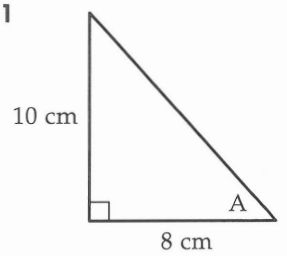


B Calculate the size of ∠A to the nearest minute.

1



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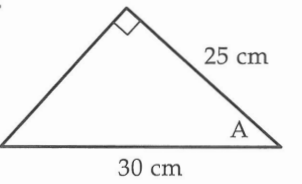

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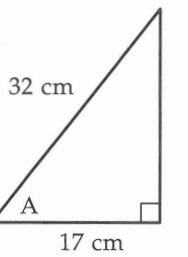

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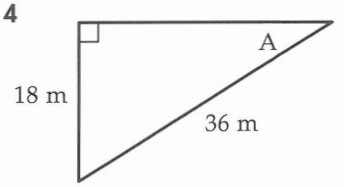

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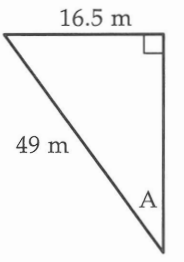

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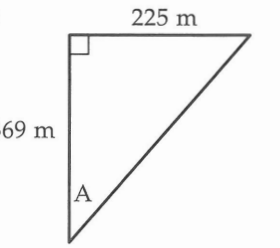

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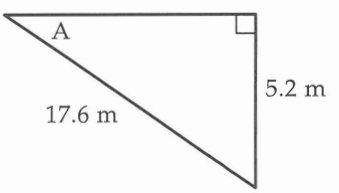

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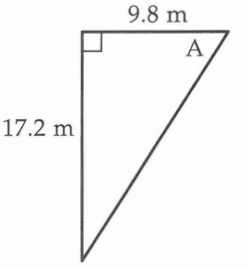

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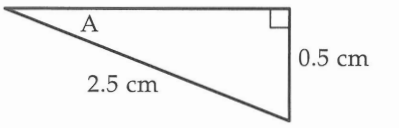

Note: diagrams not to scale

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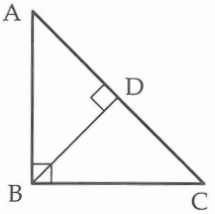
C Give all lengths correct to 1 decimal place and all angles to the nearest minute.

1 In ΔABC, $\angle B = 90°$ and $BD \perp AC$.

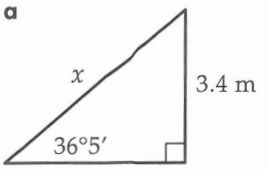
a Write 2 ratios for $\cos C$.

b Write 2 ratios for $\tan A$.



2 Determine x correct to 1 decimal place.

a



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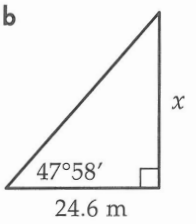

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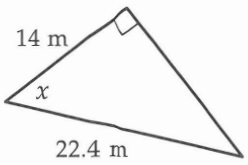

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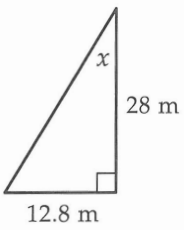
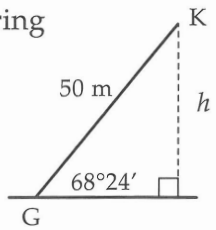
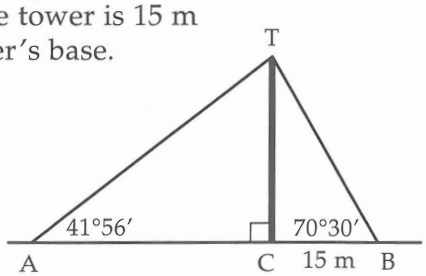


3 A kite, K, has a 50-metre string which makes an angle of 68°24′ with the ground. How high above the ground is the kite?



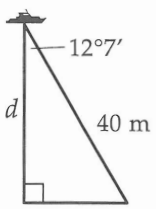
4 There are two points of observation, one on each side of a tower. Their angles of elevation to the top of the tower are 70°30′ and 41°56′. The point of observation that is closer to the tower is 15 m from the tower's base.



a Calculate the height of the tower.

b Calculate the distance between the two points of observation.

5 A ship drops a 40-metre anchor line to the ocean floor. After 2 hours the anchor line makes a 12°7′ angle to the vertical. Find the depth of the water.



6 From the top of a vertical cliff the angle of depression of a boat at sea is 42°30′. The boat is 700 m from the foot of the cliff. Calculate the height of the cliff.

7 The height of a vertical pole is 10 m and the pole's shadow is 14 m long.

a Determine the angle of elevation of the sun.

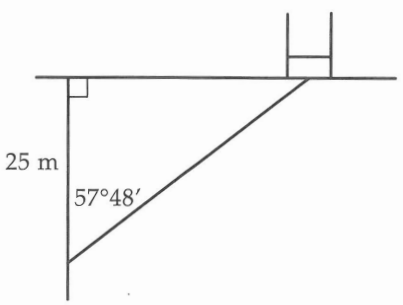
b If the shadow decreases to 5 m in length, what is the new angle of elevation of the sun?

8 From a horizontal distance of 9.2 m, the angle of elevation to the top of a building is 82°15′. Calculate the height of the building.

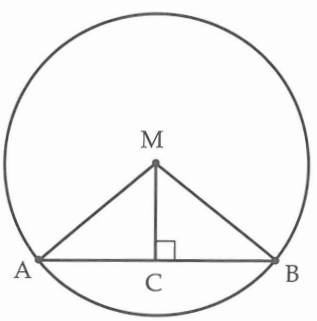
9 A boy rows a boat at an angle of 20°45′ to the river bank. After travelling 200 m he reaches the opposite bank. Calculate the width of the river.

10 A jet plane takes off at an angle of 20°59′ to the ground. It flies over a woman standing on the ground 1 km from the take-off point. How high is the jet at that point?

11 A footballer needs to kick the ball between the goalposts from the side of a rugby field. She is 25 metres from the goal line and the angle to the centre of the posts is 57°48′. Calculate the distance to the centre of the goalposts (to the nearest metre).



12 M is the centre of the circle. The chord AB measures 75.3 metres and $\angle AMB = 111°6′$. Calculate the distance MB correct to 1 decimal place. (Hint: the line MC bisects the chord AB and $\angle AMB$.)



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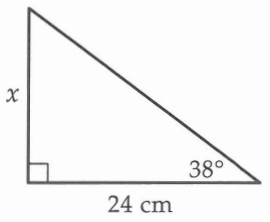
A Calculate the length of side x correct to 1 decimal place.

1


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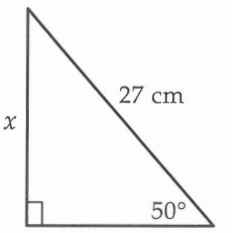

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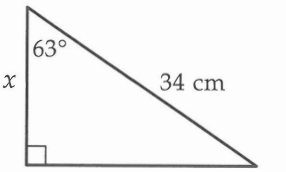

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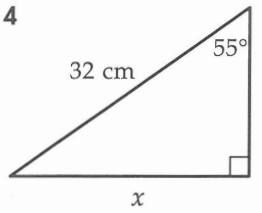

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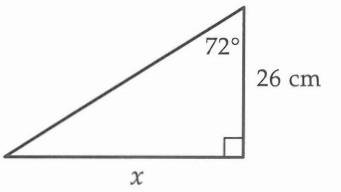

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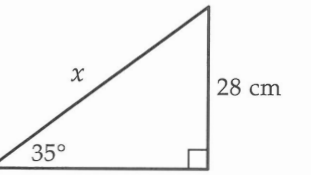
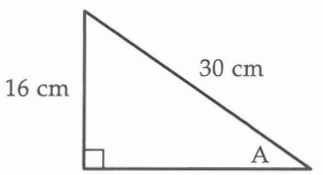


Find the size of ∠A to the nearest degree.

7


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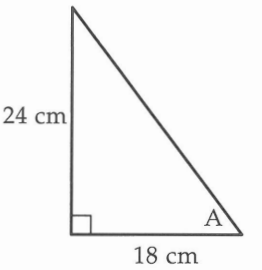

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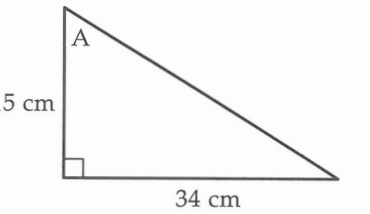

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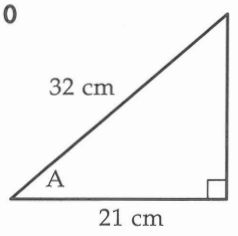

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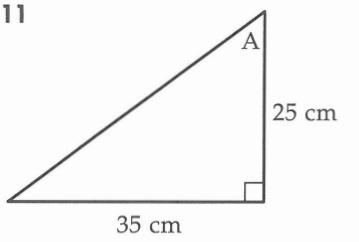

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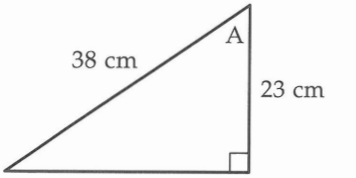

B Calculate the length of side x correct to 1 decimal place.

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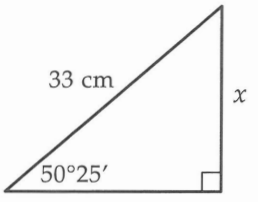

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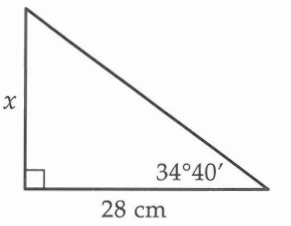

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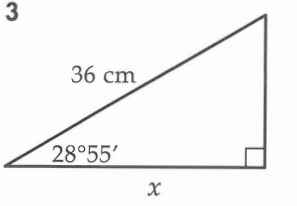

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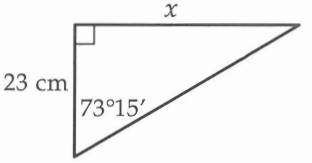

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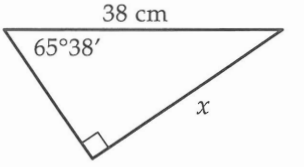

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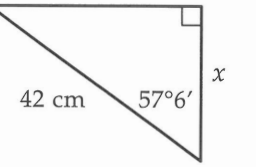

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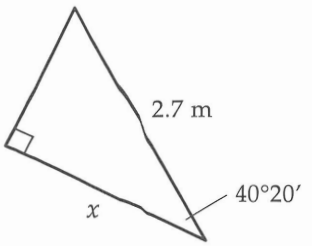
Calculate the length of side x correct to 1 decimal place.

7



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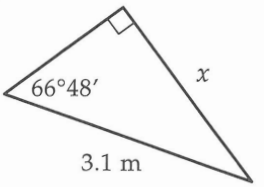

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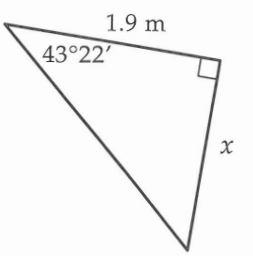
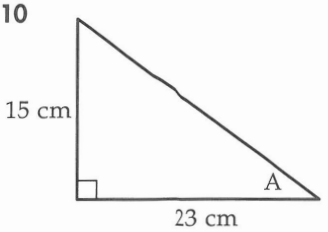


Find the size of ∠A in degrees and minutes.

10



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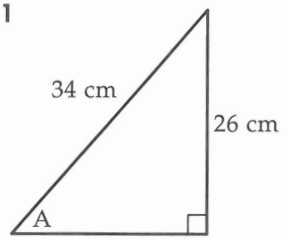

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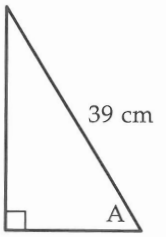

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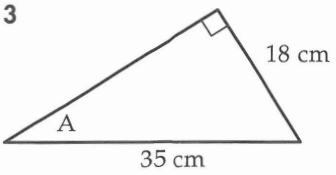

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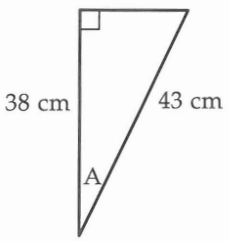

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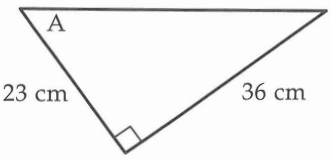
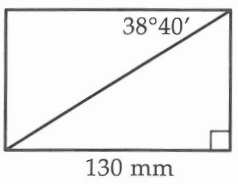
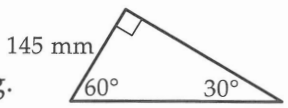


C

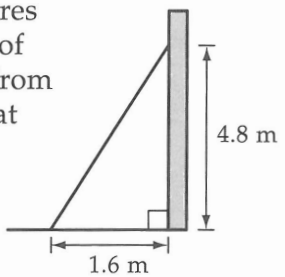
1 A rectangle is 130 mm long. The diagonal makes an angle of 38°40′ with the longer sides. Calculate the width of the rectangle to the nearest millimetre.



2 On a 30°–60° set square, the side opposite the 30° angle is 145 mm long. What is the length of the longest side to the nearest millimetre?

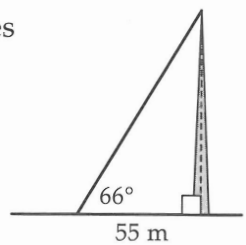


3 A ladder reaches 4.8 metres up a wall when the foot of the ladder is 1.6 metres from the base of the wall. What angle (in degrees and minutes) does the ladder make with the wall?



4 When the sun is at an angle of 48°50′ to the horizontal, a tree casts a shadow 17.8 m long. Calculate the height of the tree to 1 decimal place.

5 A girl is standing 55 metres from the base of a tower. She measures the angle of elevation of the top of the tower as 66°. Calculate the height of the tower correct to 1 decimal place.



6 A road rises 235 m in a straight length of 1.5 km. Find the road's angle of slope in degrees and minutes.

7 A small boat is 300 m from the base of a vertical cliff. The angle of depression of the boat from the top of the cliff is 23°25′. Calculate the height of the cliff to the nearest metre.

8 The vertical distance between the top and bottom of a ramp is 3.1 metres. Given that the ramp makes an angle of 14°40′ with the horizontal ground, calculate the length of the ramp correct to 1 decimal place.

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Note: diagrams not to scale

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1 The tangent ratio

A 1 a AB b AC c BC 2 a 15 cm b 8 cm c 17 cm

B 1 $\frac{3}{4}$ 2 $\frac{24}{10}=2\frac{2}{5}$ 3 $\frac{30}{40}=\frac{3}{4}$ 4 $\frac{15}{8}=1\frac{7}{8}$ 5 $\frac{20}{15}=1\frac{1}{3}$ 6 $\frac{3.2}{2.4}=\frac{4}{3}=1\frac{1}{3}$ 7 $\frac{m}{n}$ 8 $\frac{72}{135}=\frac{8}{15}$ 9 $\frac{2.5}{6}=\frac{5}{12}$ 10 $\frac{8}{6}=1\frac{1}{3}$ 11 $\frac{2}{1}=2$ 12 $\frac{20}{48}=\frac{5}{12}$ 13 $\frac{30}{40}=\frac{3}{4}$ 14 $\frac{24}{7}=3\frac{3}{7}$ 15 $\frac{y}{x}$

C 1 0.364 2 0.268 3 1.428 4 0.577 5 0.933 6 2.475 7 3.732 8 0.325 9 0.488 10 1.664 11 28.636 12 0.649 13 0.105 14 0.306 15 2.246 16 1.235 17 0.781 18 0.966 19 0.900 20 7.115 21 0.384 22 0.052 23 0.344 24 0.754 25 0.158 26 1 27 1.732 28 0.194 29 0.445 30 0.176 31 0.783 32 0.061 33 0.903 34 2.4 35 4.055 36 1.570

2 Finding the length of a side, tangent ratio

A 1 0.601 2 0.268 3 57.290 4 0.123 5 0.533 6 0.577 7 1.353 8 0.615

B 1 3.1 cm 2 1.7 m 3 47.8 cm 4 4.5 m 5 17.1 cm 6 3.2 m 7 139.3 mm 8 169.0 m 9 76.9 m 10 71.4 m 11 1.0 m 12 14.4 km 13 7.0 m 14 157.5 m 15 6.4 m

C 1 41.7 m 2 1.4 m 3 169.3 cm 4 12.1 cm 5 36.2 mm 6 22.9 m 7 30.6 m 8 9.7 m 9 13.3 cm 10 50.2 m 11 26.3 mm 12 5.0 km

3 Finding the size of an angle, tangent ratio

A 1 39° 2 33° 3 45° 4 37° 5 1° 6 88° 7 67° 8 55° 9 38° 10 23° 11 89° 12 27°

B 1 36.9° 2 45° 3 23.2° 4 67.4° 5 88.1° 6 12.7° 7 59° 8 67.9° 9 51.3° 10 41.1° 11 36.9° 12 33.7° 13 38.6° 14 33° 15 35.1°

C 1 52.1° 2 46.2° 3 18.4° 4 34.9° 5 66.1° 6 76.0° 7 65.8° 8 11.3° 9 87.2° 10 23.5° 11 25.0° 12 30° 13 30.0° 14 56.7° 15 26.6°

4 The sine ratio

A 1 13 cm 2 5 cm 3 12 cm 4 $\frac{5}{12}$ 5 17 cm 6 8 cm 7 15 cm 8 $\frac{8}{15}$

B 1 $\frac{5}{12}$ 2 $\frac{4}{10}=\frac{2}{5}$ 3 $\frac{9}{25}$ 4 $\frac{20}{24}=\frac{5}{6}$ 5 $\frac{8}{15}$ 6 $\frac{8}{17}$ 7 $\frac{15}{26}$ 8 $\frac{7}{12}$ 9 $\frac{10}{25}=\frac{2}{5}$ 10 $\frac{12}{13}$ 11 $\frac{9}{41}$ 12 $\frac{12}{15}=\frac{4}{5}$ 13 $\frac{30}{34}=\frac{15}{17}$ 14 $\frac{7}{25}$ 15 $\frac{16}{20}=\frac{4}{5}$

C 1 0.643 2 0.423 3 0.819 4 0.5 5 0.866 6 0.731 7 0.927 8 0.966 9 0.309 10 0.883 11 0.995 12 0.588 13 0.358 14 0.956 15 0.259 16 0.857 17 0.990 18 0.156 19 0.629 20 0.469 21 0.514 22 0.561 23 0.577 24 0.923 25 0.882 26 0.738 27 0.96 28 0.6 29 0.471

5 Finding the length of a side, sine ratio

A 1 $\frac{9}{16}$ 2 $\frac{14}{18}=\frac{7}{9}$ 3 $\frac{10}{30}=\frac{1}{3}$ 4 $\frac{9}{15}=\frac{3}{5}$ 5 $\frac{16}{20}=\frac{4}{5}$ 6 $\frac{7}{25}$ 7 0.699 8 0.276 9 0.469 10 0.906

B 1 9.2 cm 2 9.4 cm 3 9.2 cm 4 9.8 cm 5 16.2 cm 6 12.7 cm 7 15.1 cm 8 22.0 cm 9 28.8 cm 10 18.6 cm 11 13.0 cm 12 16.9 cm 13 15.0 cm 14 24.4 cm 15 26.3 cm

C 1 24.4 cm 2 15.2 cm 3 24.6 cm 4 20.3 cm 5 26.8 cm 6 13.7 cm 7 21.6 cm 8 23.8 cm 9 11.6 cm 10 2.9 m 11 5.5 m 12 6.5 m

6 Finding the size of an angle, sine ratio

A 1 37° 2 53° 3 30° 4 20° 5 67° 6 36° 7 15° 8 59° 9 26° 10 24° 11 49° 12 56°

B 1 25° 2 42° 3 32° 4 37° 5 44° 6 34° 7 33° 8 56° 9 60° 10 46° 11 43° 12 21°

C 1 56° 2 64° 3 45° 4 35° 5 63° 6 33° 7 62° 8 25° 9 46° 10 37° 11 23° 12 62° 13 28° 14 74° 15 53° 16 67° 17 53° 18 62° 19 30° 20 28° 21 59°

7 The cosine ratio

A 1 AB 2 AC 3 BC

B 1 $\frac{4}{5}$ 2 $\frac{5}{13}$ 3 $\frac{30}{50}=\frac{3}{5}$ 4 $\frac{8}{17}$ 5 $\frac{15}{25}=\frac{3}{5}$ 6 $\frac{3}{5}$ 7 $\frac{n}{l}$ 8 $\frac{135}{153}=\frac{15}{17}$ 9 $\frac{12}{13}$ 10 $\frac{3}{5}$ 11 $\frac{1}{\sqrt{3}}$ 12 $\frac{12}{13}$ 13 $\frac{3}{5}$ 14 $\frac{24}{25}$ 15 $\frac{x}{z}$ 16 $\frac{156}{169}=\frac{12}{13}$ 17 $\frac{4}{5}$ 18 $\frac{15}{17}$

C 1 0.940 2 0.052 3 0.999 4 0.156 5 0.719 6 0.978 7 0.259 8 0.766 9 0.934 10 0.682 11 0.809 12 0.966 13 0.574 14 0.707 15 0.407 16 0.866 17 0.999 18 0.982 19 0.225 20 0.777 21 0.921 22 0.946 23 0.500 24 0.139 25 0.469 26 0.883 27 0.515 28 0.375 29 0.946 30 0.990 31 0.998 32 0.242 33 0.602 34 0.375 35 0.191 36 0.017 37 0.799 38 0.951 39 0.906 40 0.105 41 0.292 42 0.988 43 0.105 44 0.974 45 1.000 46 0.385 47 0.385 48 0.960

8 Finding the length of a side, cosine ratio

A 1 0.866 2 0.500 3 0 4 0.707 5 0.6 6 0.866 7 0.882 8 0.923

B 1 61.8 cm 2 660.2 m 3 4.4 cm 4 97.7 m 5 1.5 cm 6 77.7 m 7 5.7 m 8 1.4 m 9 141.4 mm 10 203.2 mm 11 389.2 m 12 1 m 13 1732.1 m 14 2.2 m 15 86.7 km 16 724.6 m 17 17.5 m 18 257.4 km

C 1 18.3 m 2 1442.9 mm 3 24.1 m 4 723.4 m 5 20.1 cm 6 5.9 m 7 1123.1 km 8 16.2 m 9 15.4 m 10 22.0 m 11 4.2 km 12 10.1 m

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9 Finding the size of an angle, cosine ratio

A 1 45° 2 37° 3 60° 4 30° 5 0° 6 90° 7 76° 8 29° 9 89° 10 65° 11 55° 12 66°

B 1 23° 2 75° 3 70° 4 81° 5 53° 6 85° 7 58° 8 79° 9 78° 10 84° 11 71° 12 37° 13 36° 14 68° 15 36° 16 84° 17 31° 18 55° 19 83° 20 40° 21 79°

C 1 71.6°, 18.4° 2 59.1°, 30.9° 3 55.9°, 34.1° 4 46.8°, 43.2° 5 28.1°, 61.9° 6 70.5°, 19.5° 7 45.1°, 44.9° 8 77.8°, 12.2° 9 44.4°, 45.6° 10 82.2°, 7.8° 11 28.1°, 61.9° 12 67.4°, 22.6° 13 69.2°, 20.8° 14 67.4°, 22.6° 15 55.2°, 34.8°

10 Finding the length of a side, mixed ratios

A 1 25 cm 2 24 cm 3 7 cm 4 $\frac{24}{7} = 3\frac{3}{7}$ 5 $\frac{24}{25}$ 6 $\frac{7}{25}$ 7 $\frac{20}{29}$ 8 $\frac{21}{29}$ 9 $\frac{20}{21}$

B 1 10.2 cm 2 13.4 cm 3 18.4 cm 4 10.0 cm 5 18.0 cm 6 15.0 cm 7 20.0 cm 8 22.9 cm 9 30.1 cm 10 19.7 cm 11 20.1 cm 12 11.6 cm 13 34.3 cm 14 20.7 cm 15 22.7 cm

C 1 27.8 cm 2 15.0 cm 3 21.1 cm 4 12.4 cm 5 27.6 cm 6 38.6 cm 7 52.3 cm 8 25.5 cm 9 14.8 cm 10 1.6 m 11 1.7 m 12 2.6 m 13 1.9 m 14 1.7 m 15 1.6 m 16 1.0 m 17 1.8 m 18 2.0 m

11 Finding the size of an angle, mixed ratios

A 1 $\sin A = \frac{16}{34} = \frac{8}{17}$ $\cos A = \frac{30}{34} = \frac{15}{17}$ $\tan A = \frac{16}{30} = \frac{8}{15}$

2 $\sin A = \frac{9}{41}$ $\cos A = \frac{40}{41}$ $\tan A = \frac{9}{40}$

3 $\sin A = \frac{12}{20} = \frac{3}{5}$ $\cos A = \frac{16}{20} = \frac{4}{5}$ $\tan A = \frac{12}{16} = \frac{3}{4}$

B 1 35° 2 39° 3 32° 4 42° 5 34° 6 37° 7 35° 8 34° 9 66° 10 47° 11 51° 12 62° 13 32° 14 59° 15 48°

C 1 31° 2 40° 3 56° 4 49° 5 27° 6 58° 7 33° 8 20° 9 27° 10 32° 11 54° 12 66° 13 52° 14 26° 15 36° 16 62° 17 61° 18 40°

12 Finding the length of the hypotenuse

A 1 34 cm 2 16 cm 3 30 cm 4 20 cm 5 16 cm 6 12 cm 7 12 cm 8 16 cm

B 1 34 cm 2 26 cm 3 34 cm 4 32 cm 5 32 cm 6 31 cm 7 37 cm 8 39 cm 9 43 cm 10 29 cm 11 39 cm 12 62 cm 13 45 cm 14 52 cm 15 65 cm 16 59 cm 17 50 cm 18 41 cm

C 1 2.9 m 2 3.9 m 3 2.4 m 4 3.2 m 5 3.4 m 6 5.9 m 7 3.2 m 8 3.8 m 9 3.0 m

13 Mixed problems, right-angled triangles

A 1 20.3 cm 2 22.9 cm 3 15.5 cm 4 12.3 cm 5 29.8 cm 6 18.2 cm 7 34° 8 29° 9 62° 10 62° 11 47° 12 33°

B 1 9.4 m 2 5.7 m 3 73° 4 16.4 m 5 18° 6 92 mm 7 12 m 8 135 mm 9 38° 10 128 mm

C 1 7.3 m 2 89 mm 3 5.1 m 4 32 m 5 13.2 m 6 103 mm 7 31° 8 65.6 m 9 44° 10 55° 11 72°, 108° 12 203 mm, 435 mm

14 Angles of elevation and depression

A 1 26.8 cm 2 17.7 cm 3 29.4 cm 4 37° 5 55° 6 34°

B 1 56 m 2 84 m 3 32° 4 56 m 5 175 m 6 205 m 7 87.9 m 8 3°

C 1 16.8 m 2 41° 3 714 m 4 127.5 m 5 183 m 6 12° 7 5° 8 2682 m

15 Further problems, right-angled triangles

A 1 39.5 cm 2 49.3 cm 3 21.9 cm 4 59° 5 46° 6 45°

B 1 5° 2 30.8 m 3 24° 4 9.2 5 150 cm 6 210 cm 7 68° 8 1783 mm 9 109 mm 10 0.9 m

C 1 3.0 km 2 4.3 km 3 7.1 km 4 56.0 km 5 37 km 6 4.5 km 7 77 km 8 18.1 km 9 14.8 n. miles 10 40.9 km 11 38.2 m 12 42.4 km

16 Review A: Trigonometry, right-angled triangles

A 1 34 cm 2 16 cm 3 30 cm 4 $\frac{8}{17}$ 5 $\frac{8}{15}$ 6 $\frac{15}{17}$ 7 0.829 8 0.532 9 0.819 10 3.078

B 1 14.4 cm 2 14.1 cm 3 18.0 cm 4 17.6 cm 5 12.7 cm 6 49.5 cm 7 27.9 cm 8 39.5 cm 9 18.4 cm 10 61.9 cm 11 42.9 cm 12 47.1 cm 13 34° 14 56° 15 70° 16 40° 17 58° 18 68° 19 32° 20 53° 21 55° 22 35° 23 61° 24 27°

C 1 2.9 m 2 20.4 m 3 86 mm 4 15.7 m 5 59° 6 83.2 m 7 73 m 8 11.1 m 9 69.3 m 10 5.9 cm

Riddle: Because he wanted cold, hard cash.

17 Ratios of angles, degrees & minutes; sine cosine, tangent ratios

A 1 0.176 2 0.970 3 0.574 4 0.866 5 1 6 0.399 7 0.017 8 0.017 9 2391.831 10 0.707 11 0.707 12 1.000

B 1 1.074 2 0.692 3 0.199 4 0.803 5 3437.747 6 1.000 7 0.397 8 0.866 9 1.000 10 0.985 11 0.824 12 0.865 13 0.500 14 0.955 15 0.414 16 0.124 17 2.109 18 0.982 19 0.616 20 0.726

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21 0.982 **22** 0.340 **23** 0.127 **24** 0.429 **25** 0.019
26 0.108 **27** 0.429 **28** 1.000 **29** 0.088 **30** 0.704
31 0.471 **32** 0.224 **33** 0.704 **34** 1.793 **35** 0.794
36 1.000 **37** 0.375 **38** 0.359 **39** 0.704 **40** 0.797
41 3.130 **42** 0.500 **43** 0.367 **44** 0.706

C **Riddle:** Let us roar together.

18 *Finding lengths of sides, angles in degrees & minutes*

A **1** 0.4 **2** 0.2 **3** 0.2 **4** 0.5 **5** 1.0 **6** 75.2 m **7** 63.6 **8** 14.4 **9** 6.6

B **1** 5.3 m **2** 997.6 m **3** 2.8 m **4** 9.1 m **5** 450.3 m **6** 116.0 km **7** 15.0 m **8** 0.1 m **9** 56.2 km **10** 34.8 km **11** 6.7 m **12** 104.9 m **13** 22.1 m **14** 3768.2 mm **15** 26.2 km **16** 105.0 cm **17** 12.7 m **18** 2.1 km

C **1** 28.28 m **2** 37.24 m **3** 15.47 m **4** 36.56 m **5** 55.20 m **6** 206.88 m **7** 128.98 m **8** 2.54 m **9** 54.31 m

19 *Finding sizes of angles, degrees & minutes*

A **1** 39° **2** 45° **3** 48° **4** 29° **5** 25° **6** 16° **7** 79° **8** 25°

B **1** 79°24′ **2** 13°2′ **3** 88°34′ **4** 23°55′ **5** 75°23′ **6** 36°31′ **7** 69°38′ **8** 30°23′ **9** 14°29′ **10** 77°38′ **11** 63°45′ **12** 83°35′ **13** 44°54′ **14** 62°44′ **15** 56°57′ **16** 41°30′ **17** 45°6′ **18** 26°34′

C **1** 72°44′ **2** 75°12′ **3** 57°16′ **4** 34°31′ **5** 35°21′ **6** 36°25′ **7** 34°42′ **8** 60°40′ **9** 22°8′ **10** 5°1′ **11** 62°49′ **12** 18°43′

20 *Mixed problems, angles in degrees & minutes*

A **1** 10.5 cm **2** 15.0 m **3** 77.1 cm **4** 9.2 m **5** 34.2 m **6** 5.8 m **7** 350.2 cm **8** 351.6 m **9** 17.3 m

B **1** 51°20′ **2** 33°33′ **3** 57°55′ **4** 30° **5** 19°41′ **6** 31°22′ **7** 17°11′ **8** 60°20′ **9** 11°32′

C **1 a** $\frac{BC}{AC}$ $\frac{DC}{BC}$ **b** $\frac{BC}{AB}$ $\frac{BD}{AD}$ **2 a** 5.8 m **b** 27.3 m **c** 51°19′ **d** 24°34′ **3** 46.5 m **4 a** 42.4 m **b** 62.2 m **5** 39.1 m **6** 641.4 m **7** 35°32′, 63°26′ **8** 67.6 m **9** 70.9 m **10** 0.4 km **11** 47 m **12** 45.7 m

21 *Review B: Trigonometry, right-angled triangles*

A **1** 18.8 cm **2** 20.7 cm **3** 15.4 cm **4** 26.2 cm **5** 80.0 cm **6** 48.8 cm **7** 32° **8** 53° **9** 64° **10** 49° **11** 54° **12** 53°

B **1** 25.4 cm **2** 19.4 cm **3** 31.5 cm **4** 76.4 cm **5** 34.6 cm **6** 22.8 cm **7** 2.1 m **8** 2.8 m **9** 1.8 m **10** 33°7′ **11** 49°53′ **12** 68°58′ **13** 30°57′ **14** 27°54′ **15** 57°26′

C **1** 104 mm **2** 290 mm **3** 18°26′ **4** 20.4 m **5** 123.5 m **6** 9°1′ **7** 130 m **8** 12.2 m

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